

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY,
BELGAUM, KARNATAKA**



PROJECT REPORT

ON

“AI - Powered Placement Preparation Platform”

Submitted in partial fulfillment of the requirement for the award of the degree of

**BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING**

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CERTIFICATE

Certified that the project work and presentation entitled “AI- Powered Placement Preparation Platform” is a bonafide work carried out by Neha Vithob Nayak (2SD22CS129), Pooja S D (2SD22CS130), Ambika Chavan (2SD23CS400), Smital Kaginkar (2SD23CS409), students of S. D. M. College of Engineering & Technology, Dharwad, in partial fulfillment for the award of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belgaum, during the year 2025-2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Project has been approved, as it satisfies the academic requirements in respect of project report prescribed for the said degree.

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ABSTRACT

In today's highly competitive job market, students must develop strong technical, aptitude, and communication skills to succeed in campus placements. However, existing preparation methods are often fragmented, relying on multiple platforms, unstructured study materials, and limited personalized guidance. This lack of a unified system makes it difficult for students to track progress, identify weaknesses, and prepare effectively for recruitment processes.

*To overcome these challenges, the **AI Powered Placement Preparation Platform** is proposed as an integrated and intelligent web-based system designed to streamline and enhance the placement preparation process. The platform combines multiple essential modules, including aptitude practice, coding assessments, resume analysis, and AI-driven mock interviews, into a single centralized environment.*

The system leverages Artificial Intelligence techniques such as machine learning and natural language processing to deliver personalized learning experiences. It analyzes user performance to recommend suitable questions, adapts difficulty levels dynamically, and provides detailed feedback. The resume analysis module evaluates resumes against industry standards and suggests improvements, while the AI mock interview module simulates real-world interview scenarios and offers constructive feedback to improve communication and confidence.

Additionally, the platform includes performance tracking dashboards that allow students to monitor their progress and identify areas requiring improvement. The system is developed using modern web technologies, ensuring scalability, responsiveness, and ease of use.

By integrating intelligent automation with structured learning resources, the proposed platform provides a comprehensive solution for placement preparation. It reduces reliance on scattered resources and equips students with the skills and confidence required to meet industry expectations, thereby improving overall placement success rates.

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PROBLEM STATEMENT

AI- Powered Placement Preparation Platform

Develop a comprehensive and intelligent web-based platform that assists students in preparing for campus placements by integrating aptitude practice, coding assessments, resume analysis, and AI-driven mock interviews. The system should provide personalized feedback, performance tracking, and adaptive learning to improve student readiness and placement success rates.

CHAPTER 1

INTRODUCTION

In today's highly competitive job market, securing placement opportunities requires more than just academic knowledge. Students must be proficient in aptitude tests, technical coding skills, resume building, and interview performance. However, most students rely on scattered resources such as different websites, online videos, and practice platforms, which lack structure and personalized guidance.

This fragmented approach leads to inefficient preparation, lack of progress tracking, and limited feedback on performance. Students often struggle to identify their weak areas, and there is no unified system that simulates real placement scenarios or provides data-driven insights.

To address these challenges, the **AI Powered Placement Preparation Platform** is proposed as a centralized and intelligent solution. The platform integrates multiple preparation modules such as aptitude tests, coding practice, resume evaluation, and AI-based mock interviews into a single system.

Using Artificial Intelligence techniques, the platform analyzes user performance, provides personalized recommendations, adapts question difficulty, and generates feedback. The system also tracks progress through dashboards, enabling students to continuously improve their skills.

By combining structured learning with intelligent automation, the platform aims to bridge the gap between academic preparation and industry expectations, thereby enhancing placement readiness and success.

CHAPTER 2

SUSTAINABLE DEVELOPMENT GOALS (SDG)

The United Nations' 17 Sustainable Development Goals (SDGs) provide a comprehensive framework that increasingly shapes how engineering projects are conceived, designed, and executed worldwide. Engineers play a pivotal role in achieving these goals, as infrastructure, technology, and innovation are fundamental enablers of sustainable development. From SDG 6 (Clean Water and Sanitation) driving water treatment facility designs to SDG 7 (Affordable and Clean Energy) influencing renewable energy projects, these goals are no longer peripheral considerations but central to project planning and evaluation. Modern engineering projects are expected to demonstrate clear alignment with relevant SDGs, often requiring environmental impact assessments, social benefit analyses, and long-term sustainability metrics that extend beyond traditional cost-benefit calculations.

Engineering projects now commonly integrate multiple SDGs simultaneously, recognizing the interconnected nature of sustainable development. For instance, a transportation infrastructure project might address SDG 9 (Industry, Innovation and Infrastructure) while also contributing to SDG 11 (Sustainable Cities and Communities) through reduced emissions and improved urban mobility, and SDG 13 (Climate Action) through low-carbon construction methods. This multi-dimensional approach requires engineers to adopt systems thinking, considering not just technical performance but also social equity, environmental protection, and economic viability throughout a project's lifecycle.

The SDG framework has transformed engineering practice by establishing measurable targets and indicators that help quantify project impacts beyond traditional engineering metrics. Projects are increasingly evaluated on their contribution to goals like SDG 3 (Good Health and Well-being), SDG 12 (Responsible Consumption and Production), and SDG 15 (Life on Land), requiring engineers to collaborate with diverse stakeholders including environmental scientists, social planners, and community representatives. This paradigm shift has elevated the profession's responsibility, positioning engineers as key agents of sustainable development who must balance technical excellence with ethical considerations, resource efficiency, and long-term societal benefit in every project they undertake.

The proposed system aligns with global sustainable development initiatives by contributing to education, innovation, and employability. The **AI Powered Placement Preparation Platform** primarily supports **SDG 4: Quality Education** by providing structured and accessible placement preparation resources with personalized and continuous learning beyond traditional classrooms. It also contributes to **SDG 8: Decent Work and Economic Growth** by enhancing student employability and improving placement success rates, thereby supporting better career opportunities. Furthermore, the platform supports **SDG 9: Industry, Innovation, and Infrastructure** by leveraging modern technologies such as Artificial Intelligence, web applications, and data analytics to create a scalable and innovative digital infrastructure for education. Overall, the system promotes skill development, technological advancement, and sustainable career growth.

CHAPTER 3

OBJECTIVES

1. To provide adaptive aptitude and coding practice based on student performance and evolving placement trends.
2. To evaluate resumes using AI and Natural Language Processing techniques to provide structured improvement suggestions.
3. To conduct AI-based mock video interviews and provide structured performance feedback on communication, confidence, and technical responses.
4. To perform analysis of student skills, academic records, and career goals in order to derive comprehensive performance statistics and system-wide insights.

CHAPTER 4

LITERATURE REVIEW

S.No	Title & Authors	Publication	Methodology	Limitations
1	Smart Resume Analyzer – Sharma et al.	IEEE 2021	NLP-based resume keyword extraction	No personalized recommendations
2	AI-Based Student Performance Prediction – Kumar et al.	Springer 2020	Machine Learning classification models	Not placement-focused
3	Online Coding Evaluation System – Rao et al.	IJCA 2019	Automated test case execution	No analytics or AI guidance
4	Placement Management System – XYZ University	2022	CRUD-based portal	No AI integration
5	Adaptive Learning System – MIT Research	2021	Recommendation algorithm	Complex implementation, not campus-specific

Existing systems lack integration of:

- Resume analysis
- Adaptive testing
- AI video interview evaluation
- Placement readiness scoring
- Institutional analytics

Our project integrates all these components into a single intelligent ecosystem.

CHAPTER 5

DETAILED DESIGN

1. SYSTEM ARCHITECTURE:

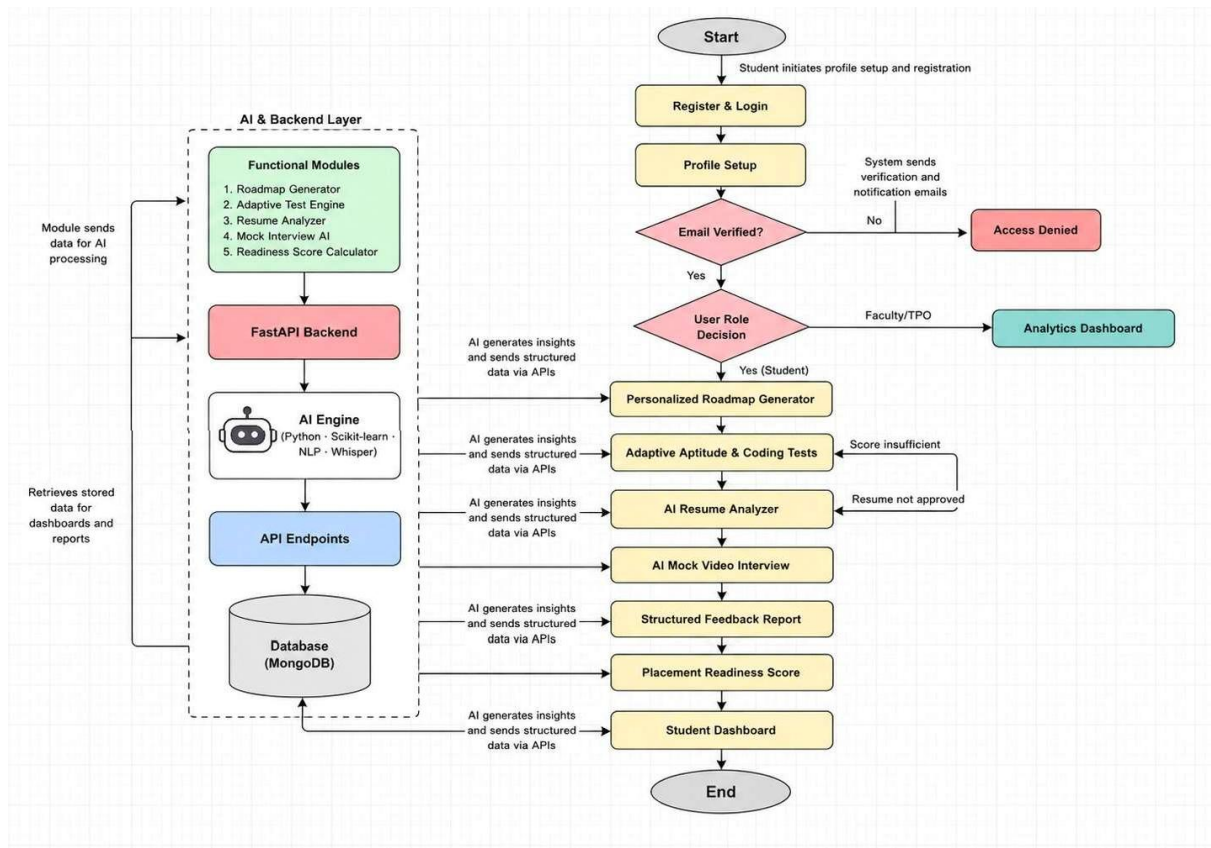


Figure 1. System Architecture

The system architecture of the **AI Powered Placement Preparation Platform** is designed using a modular and intelligent layered approach to provide personalized placement preparation and real-time performance analysis. The architecture integrates frontend workflows, backend services, AI modules, and database management to ensure scalability, efficiency, and secure data handling.

The process begins when a student accesses the platform and completes registration and profile setup. After registration, the system performs email verification to authenticate the user. If verification fails, access is denied; otherwise, the user proceeds based on role selection. Faculty members and Training & Placement Officers (TPOs) are redirected to the analytics dashboard, while students are directed to the personalized preparation modules.

The platform includes multiple AI-driven functional modules such as the **Roadmap Generator**, **Adaptive Test Engine**, **Resume Analyzer**, **AI Mock Interview System**, and **Readiness Score Calculator**. These modules interact with the backend through REST APIs and process user data

intelligently to generate structured insights and recommendations.

The backend is implemented using **FastAPI**, which handles API communication, request processing, authentication, and integration with AI services. The AI engine is developed using Python-based technologies such as **Scikit-learn**, **Natural Language Processing (NLP)**, and **Whisper AI**, enabling functionalities like resume analysis, adaptive testing, interview evaluation, and feedback generation.

The system stores all user information, test results, interview reports, and analytics data in a **MongoDB database**, ensuring efficient retrieval and scalability. The AI modules continuously retrieve stored data, analyze user performance, and generate personalized outputs such as learning roadmaps, structured feedback reports, placement readiness scores, and performance dashboards.

The workflow for students includes personalized roadmap generation, adaptive aptitude and coding assessments, AI-based resume analysis, AI mock video interviews, and detailed performance evaluation. Based on the generated scores and analysis, the system provides improvement suggestions and continuously updates the student dashboard. This architecture enables intelligent automation, personalized learning, and data-driven placement preparation within a centralized platform.

2. USE CASE DIAGRAM:



Figure 2. Use Case diagram

The use case diagram of the **AI Powered Placement Preparation Platform** illustrates the interactions between the three primary actors of the system: **Student, Faculty, and TPO (Training and Placement Officer)**. The diagram represents the functionalities accessible to each user role and demonstrates how the platform supports placement preparation, performance monitoring, and institutional analytics.

The **Student** is the primary user of the platform and can perform several activities related to placement preparation. Students can register and log in to the system securely, manage their profiles, and generate personalized preparation roadmaps based on their skills and performance. The platform allows students to attempt aptitude and coding tests, upload resumes for AI-based analysis, and attend AI-powered mock interviews. After completing these activities, students can view their placement readiness score and monitor their progress through a performance dashboard that displays test scores, feedback, and improvement areas.

The **faculty** actor is responsible for monitoring and evaluating student progress. Faculty members can log in to the system and access features such as viewing student performance, analyzing department-wise placement readiness, and generating performance reports. These functionalities help faculty track student development and provide guidance for improving placement outcomes.

The **Training and Placement Officer (TPO)** play an administrative and analytical role within the system. The TPO can log in to access institutional-level analytics, monitor overall placement readiness, and download placement reports for evaluation purposes. Additionally, the TPO can send job alerts and placement notifications to students, ensuring effective communication regarding recruitment opportunities and placement activities.

Overall, the use case diagram demonstrates how the platform integrates AI-driven placement preparation with analytics and administrative functionalities, enabling students, faculty, and placement officers to collaborate effectively within a centralized intelligent system.

3. DATA FLOW DIAGRAM:



Figure 3. Data Flow Diagram

The **AI Powered Placement Preparation Platform** is designed to provide an intelligent and personalized placement preparation experience using Artificial Intelligence and modern web technologies. The system begins with smart onboarding, where students enter their academic details and skills. Based on this information, the AI generates personalized preparation roadmaps and provides modules such as aptitude tests, coding challenges, resume analysis, and AI-based mock interviews. The platform continuously evaluates student performance using adaptive testing and calculates a Placement Readiness Score (PRS) to identify strengths and improvement areas. Performance analytics and reports are then generated for students, faculty members, and TPOs.

The technical architecture consists of a frontend developed using **React.js**, a backend implemented with **FastAPI**, and a **MongoDB** database for data storage. Secure authentication is handled using JWT and bcrypt. The AI engine performs resume analysis, speech evaluation, interview feedback generation, and personalized recommendations using NLP and machine learning techniques. The system also includes notification and file storage modules to support real-time updates and secure resume uploads. Overall, the architecture provides a scalable, secure, and AI-driven ecosystem for improving placement readiness.

4. ER DIAGRAM:

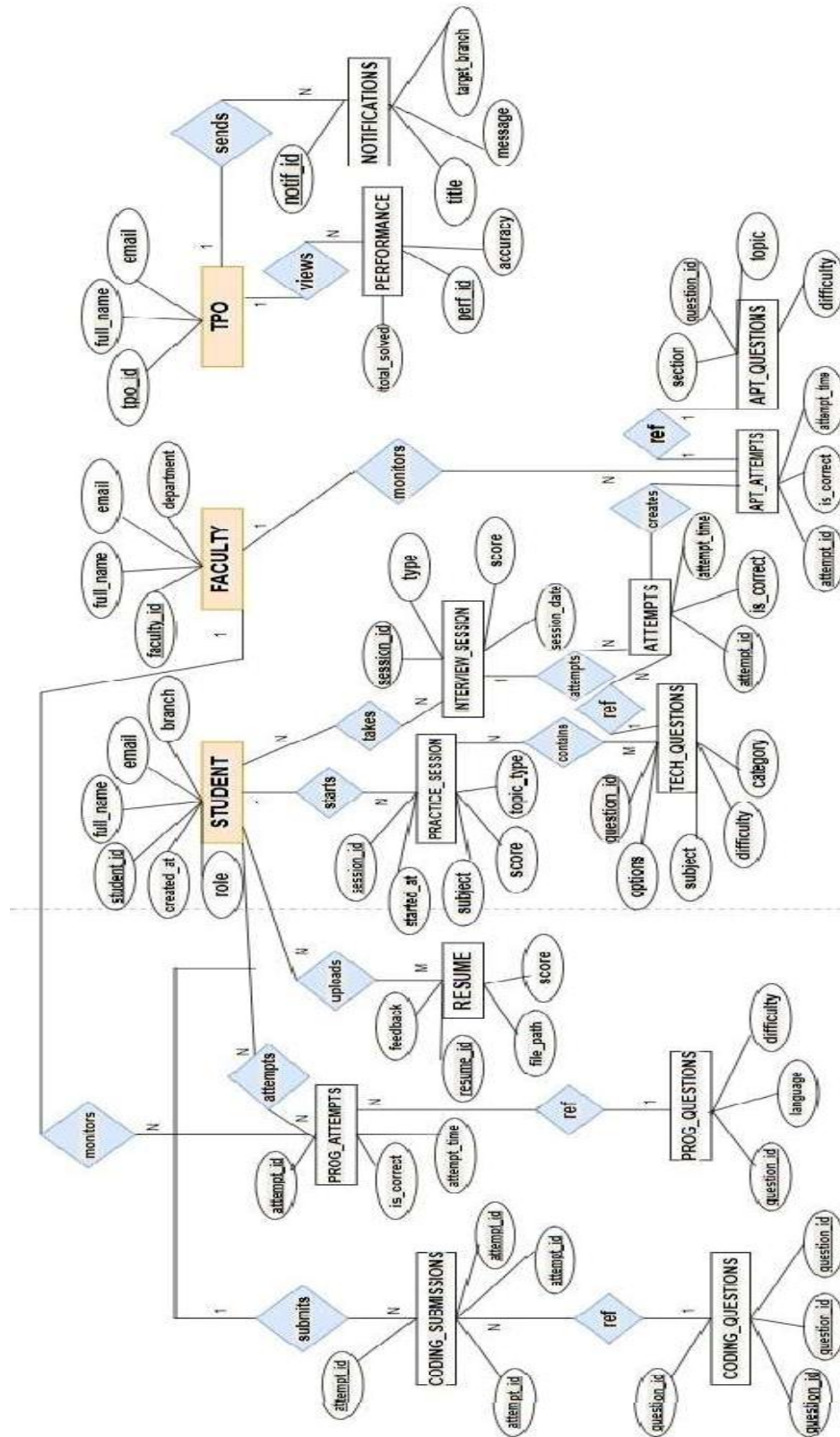


Figure 4. ER diagram

5. CLASS DIAGRAM:

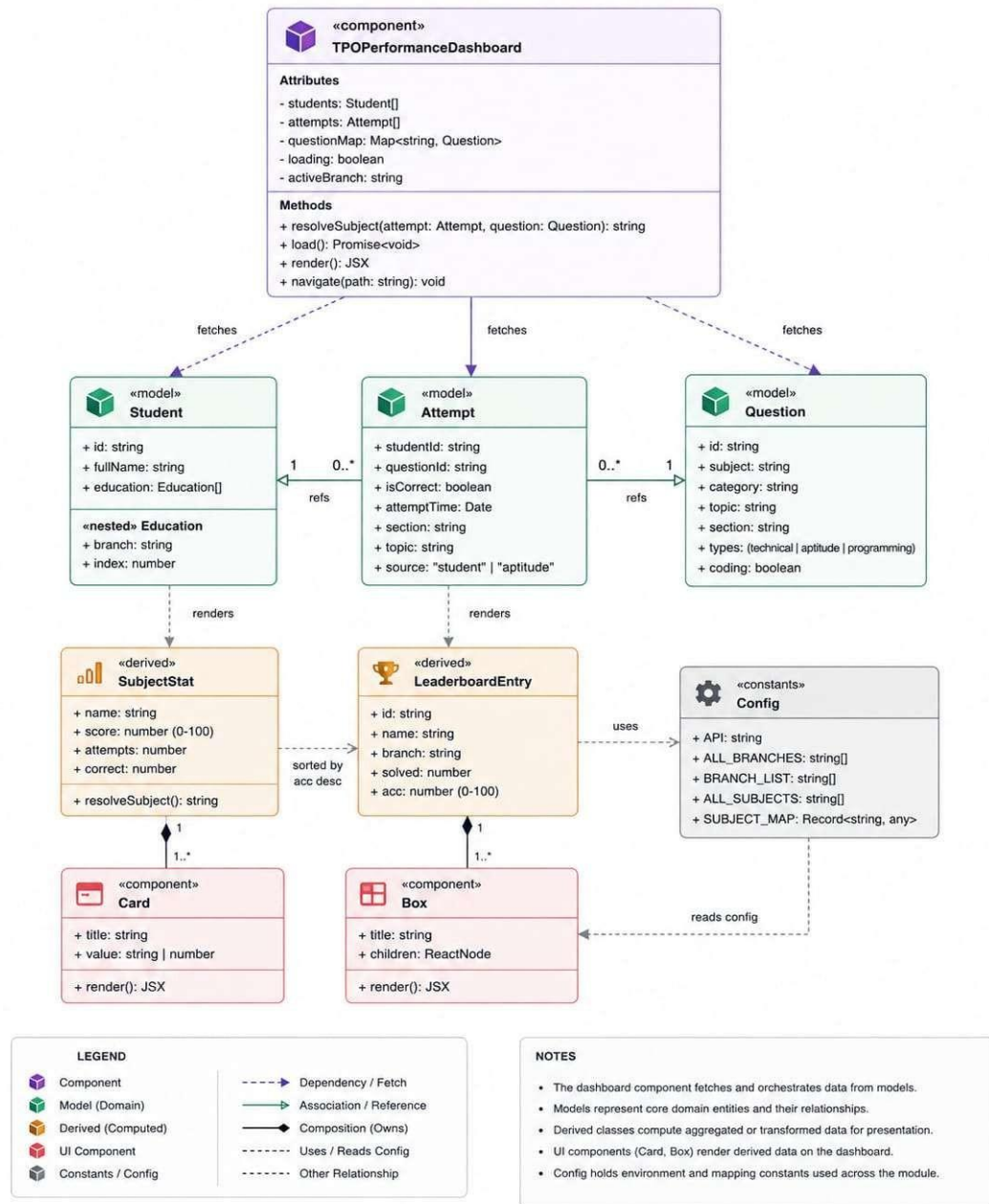


Figure 5. Class Diagram

The class diagram represents the **TPO Performance Dashboard** module of the **AI Powered Placement Preparation Platform**. The main component, **TPOPerformanceDashboard**, manages and fetches data from core models such as **Student**, **Attempt**, and **Question** to track student performance and assessment activities. These models store details related to students, test attempts, and questions. The system also includes derived classes like **SubjectStat** and **LeaderboardEntry** to generate analytics such as scores, accuracy, and rankings. UI components like **Card** and **Box** display the analytics visually, while the **Config** class stores configuration details such as API links and subject mappings. Overall, the diagram shows how data, analytics, and UI components work together to provide performance monitoring and placement analytics.

6. ACTIVITY DIAGRAM

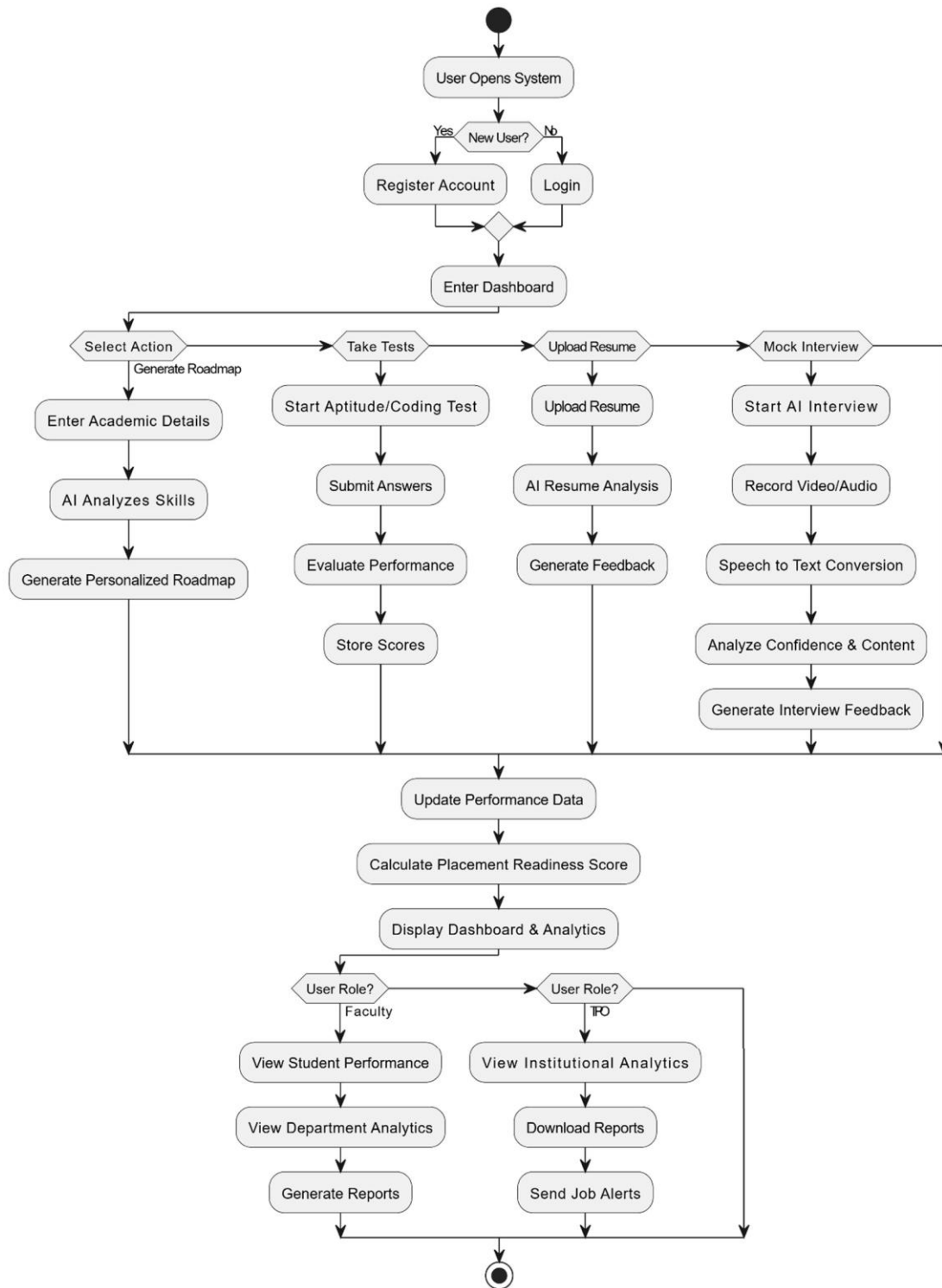


Figure 6. Activity diagram

The activity diagram of the **AI Powered Placement Preparation Platform** illustrates the complete workflow of the system, starting from user access to performance analysis and reporting. It represents the sequence of activities performed by students, faculty members, and Training and Placement Officers (TPOs) within the platform.

The process begins when the user opens the system. The system first checks whether the user is a new user or an existing user. New users are required to register an account, while existing users can directly log in using their credentials. After successful authentication, the user enters the main dashboard of the platform.

Inside the dashboard, the student can select different placement preparation activities. One of the primary functionalities is the **Roadmap Generation Module**, where students enter their academic details and skill information. The AI engine analyzes these inputs and generates a personalized preparation roadmap based on the student's strengths, weaknesses, and career goals.

Another important module is the **Test Module**, where students can attempt aptitude and coding tests. After starting the test, the student submits answers, and the system evaluates performance automatically. The obtained scores are then stored in the database for future analysis and progress tracking.

The platform also provides an **AI Resume Analysis Module**. In this process, students upload their resumes, which are analyzed using Artificial Intelligence and Natural Language Processing techniques. The system evaluates the resume content, formatting, skills, and keywords, and then generates feedback and improvement suggestions.

The **AI Mock Interview Module** simulates real interview scenarios. Students participate in AI-driven mock interviews where their video or audio responses are recorded. The system converts speech to text and analyzes factors such as confidence, communication skills, and content quality. Based on this analysis, detailed interview feedback is generated to help students improve their performance.

After all activities are completed, the system updates the performance database and calculates the student's **Placement Readiness Score**. The platform then displays dashboards and analytics containing test performance, interview analysis, resume feedback, and overall readiness metrics.

Finally, based on the user role, the workflow diverges into administrative functionalities. Faculty members can view student performance, department analytics, and generate reports, while TPOs can access institutional analytics, download placement reports, and send job alerts or notifications to students. The process concludes after all required operations and analyses are completed successfully.

7. SEQUENCE DIAGRAM:

a) Authentication Module

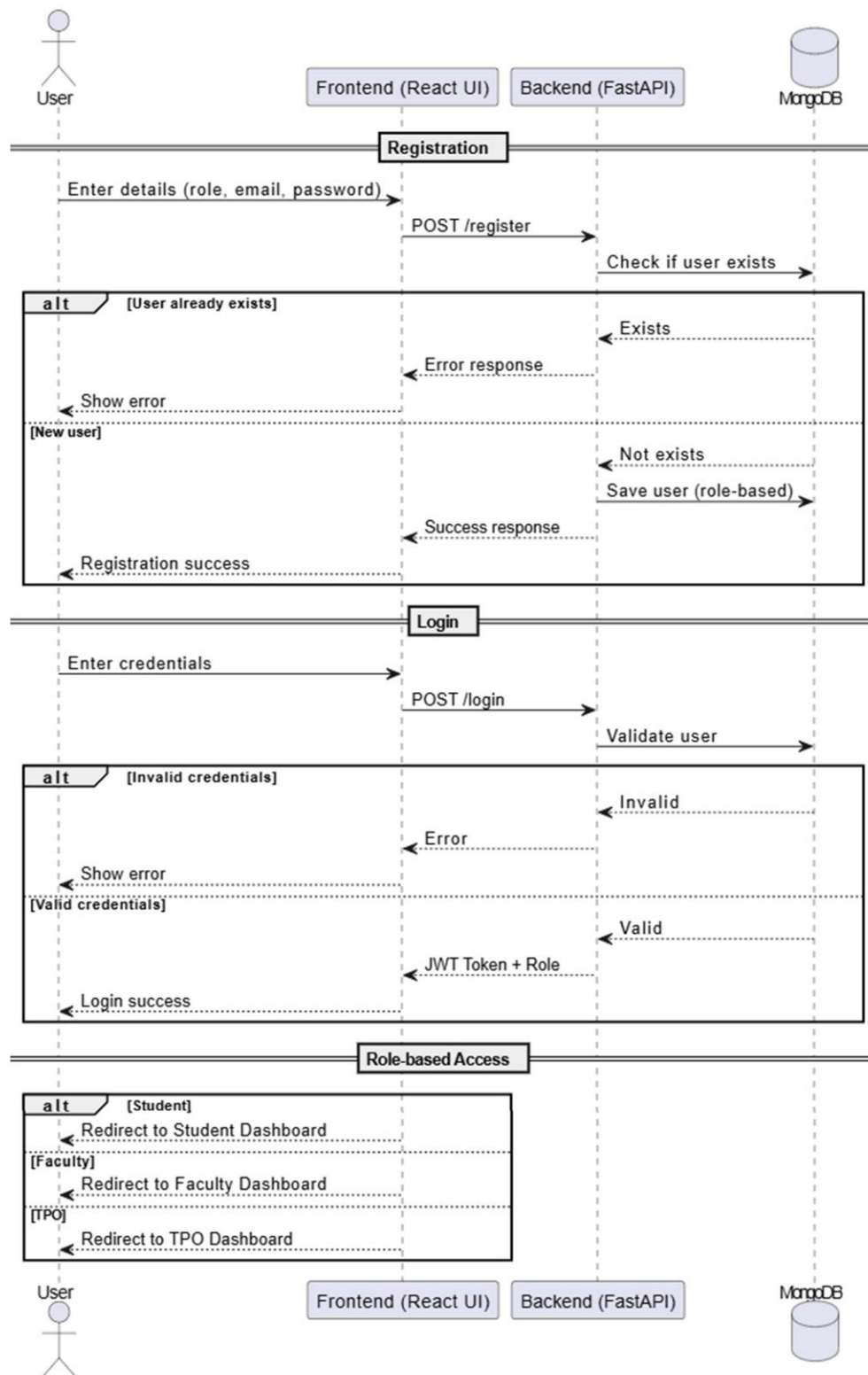


Figure 7. Sequence diagram for Authentication Module

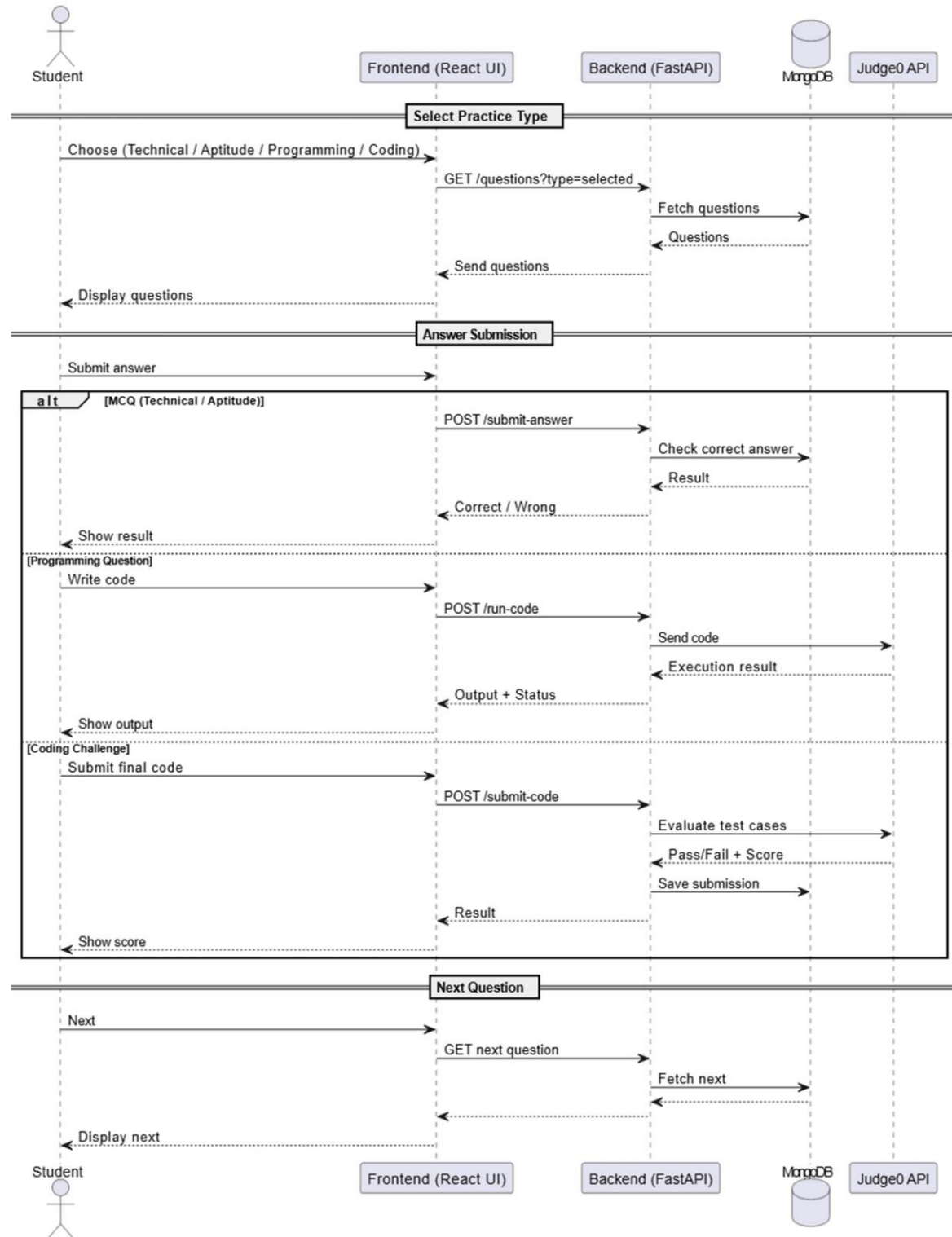
b) Practice Module (Coding and Aptitude)

Figure 8. Sequence diagram for Practice Module

c) Mock Interview Module

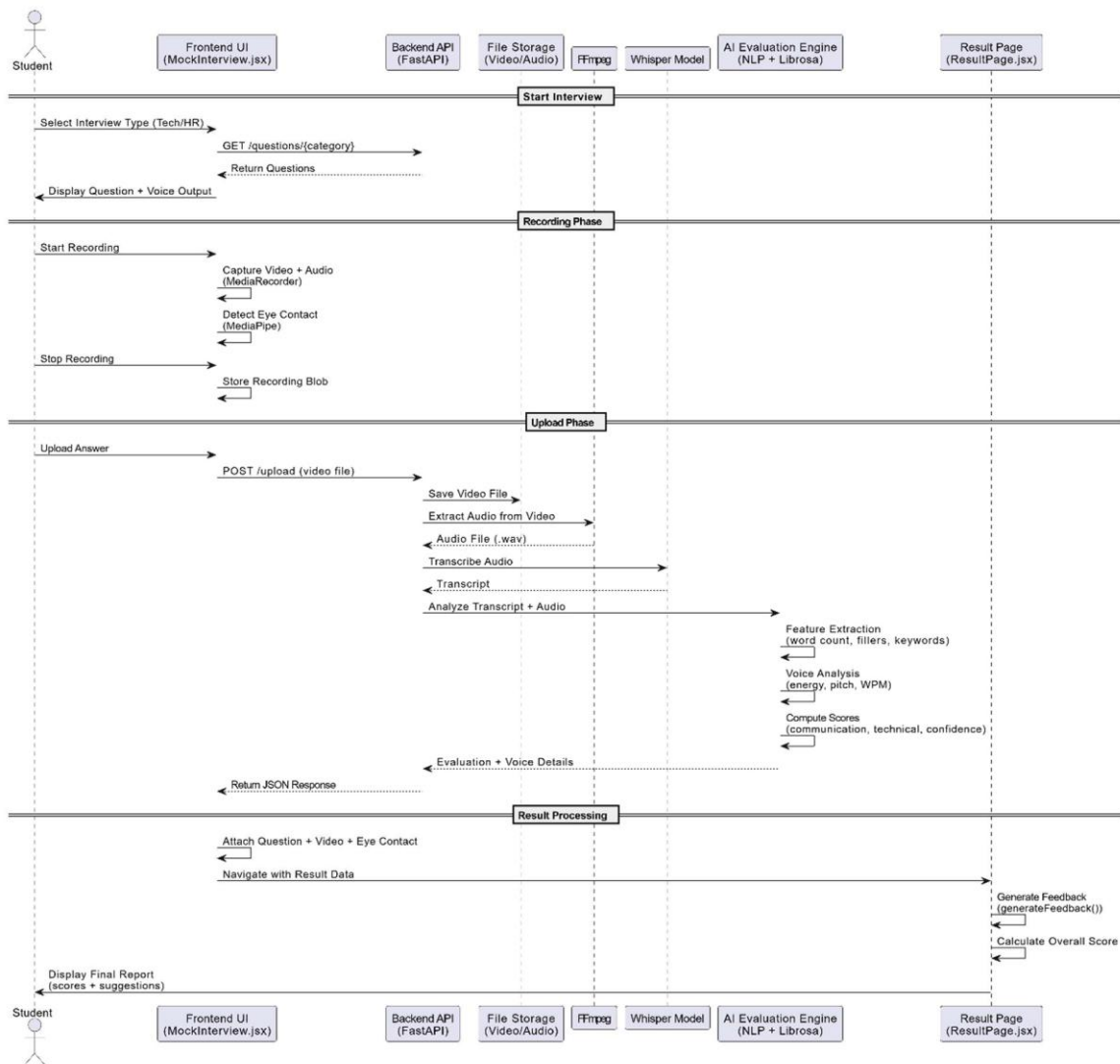


Figure 9. Sequence diagram for Mock Interview module

d) Resume Analyzer Module

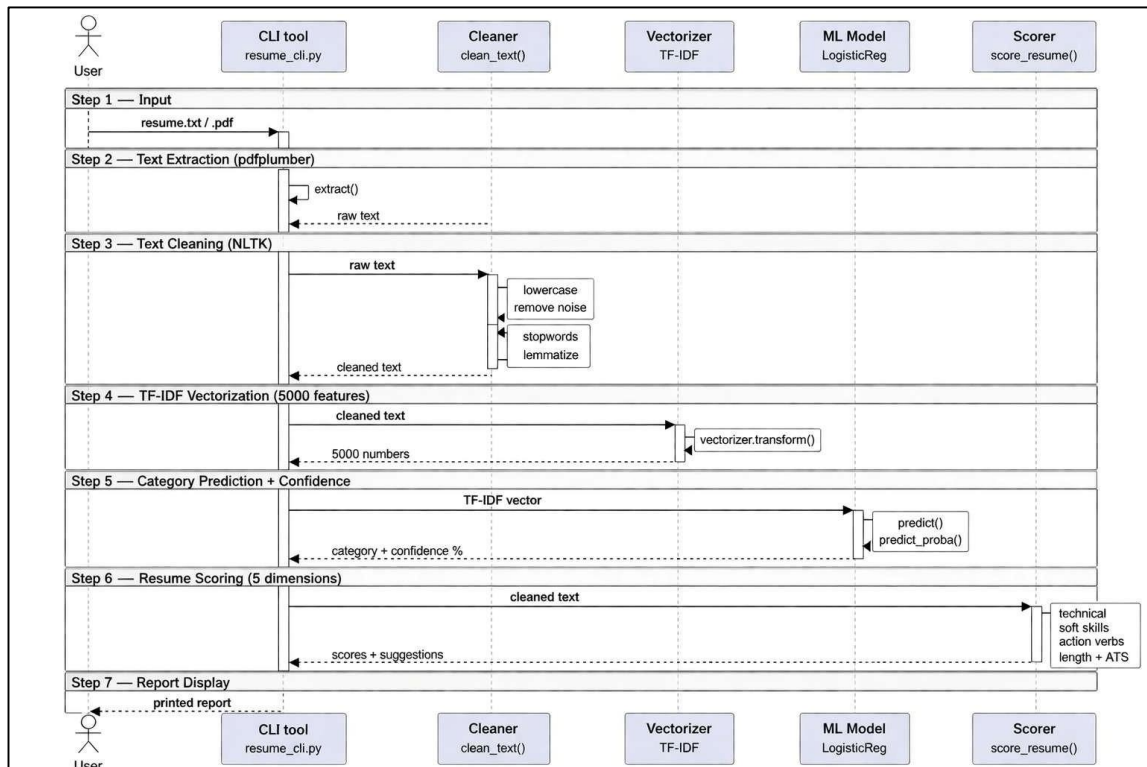


Figure 10. Sequence diagram for Resume Analyzer Module

e) Notification Module

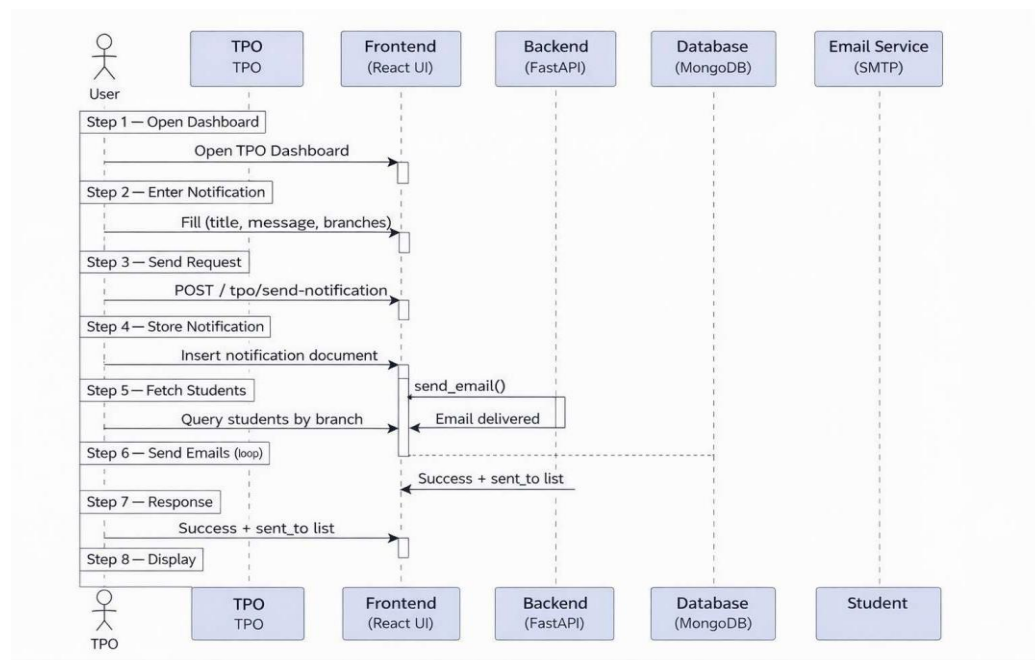


Figure 11. Sequence diagram for Notification Module

CHAPTER 6

PROJECT SPECIFIC REQUIREMENTS

Hardware Requirements

- Multi-core processor with minimum 8 GB RAM and SSD storage for server-side processing and AI model execution.
- Client systems should support modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge.
- Mobile compatibility with Android and iOS devices for responsive access to the platform.
- Stable internet connection with secure HTTPS communication for uninterrupted access to AI services and placement modules.
- Webcam and microphone support for AI-based mock interview sessions.

Software Requirements

- **Frontend:** React.js
- **Backend:** FastAPI (Python)
- **Database:** MongoDB (NoSQL Database)
- **AI & Machine Learning:** Python, Scikit-learn, NLP libraries, Whisper AI
- **Development IDE:** Visual Studio Code
- **Version Control:** Git and GitHub
- **Design Tools:** Draw.io for UML/ER diagrams, Figma or Stitch for UI design
- **Authentication:** JWT-based secure authentication system
- **APIs:** RESTful APIs for frontend-backend communication
- **Testing & Debugging Tools:** Postman and Thunder Client
- **Deployment Tools:** Vercel (optional)

Security & Authentication Requirements

- Only authenticated users with valid institutional credentials can access the platform.
- Secure JWT tokens are used for maintaining login sessions and role-based access.
- All sensitive user data and credentials are encrypted and securely stored.
- Data transmission between client and server is protected using HTTPS protocol.
- Passwords are securely hashed before storage in the database.
- AI-generated reports and interview data are accessible only to authorized users.

Functional Requirements

1. User Registration and Authentication

- Students, faculty, and TPOs must be able to register and log in securely.
- JWT-based authentication should provide secure session management.

2. User Profile Management

- Users should be able to create, view, and update their profiles.
- Students can add academic details, technical skills, and career interests.

3. Personalized Roadmap Generation

- The system should generate AI-based preparation roadmaps according to student performance, skills, and career goals.
- Roadmaps must dynamically update based on progress and assessment scores.

4. Aptitude and Coding Test Module

- Students can attempt aptitude and coding tests.
- The system evaluates answers automatically and stores performance data.
- Adaptive testing should adjust difficulty levels based on user performance.

5. Resume Analysis Module

- Students can upload resumes in supported formats.
- AI-based analysis should evaluate resume structure, keywords, skills, and formatting.
- The system should generate feedback and improvement suggestions.

6. AI Mock Interview Module

- Students can participate in AI-powered mock interviews.
- The system should record audio/video responses and convert speech to text.
- AI should analyze communication skills, confidence, and answer quality.
- Detailed interview feedback should be generated automatically.

7. Placement Readiness Score

- The system should calculate an overall placement readiness score based on tests, interviews, and resume analysis.
- Scores should update dynamically after every activity.

8. Dashboard and Analytics

- Students should view their performance analytics and improvement areas.
- Faculty and TPOs should access department-level and institutional placement analytics.
- Reports should be downloadable for analysis and monitoring.

9. Notification System

- Users should receive notifications regarding test schedules, interview feedback, and placement updates.
- Notifications must be displayed in real-time.

10. Search and Recommendation System

- Users should be able to search for topics, tests, and preparation resources.
- AI should recommend learning resources and practice modules based on performance analysis.

Non-Functional Requirements

1. Performance

The platform should provide fast response times, with dashboards loading within 2 seconds and AI analysis modules generating feedback efficiently.

2. Scalability

The system should support increasing numbers of students, faculty members, and AI processing requests without performance degradation.

3. Availability

The platform should remain accessible to authorized users at all times with minimal downtime.

4. Security

All user information, interview recordings, and performance data must be securely stored and transmitted using encrypted communication protocols.

5. Maintainability

The application should follow a modular architecture using React.js, Node.js, and AI service separation for easier maintenance and future upgrades.

6. Usability

The platform should provide a clean, user-friendly, and responsive interface that allows students to easily navigate preparation modules and analytics dashboards.

7. Reliability

The system should consistently provide accurate AI-based evaluations, recommendations, and analytics without data loss or processing failures.

8. Portability

The platform should operate smoothly across different operating systems, browsers, and mobile devices without compatibility issues.

CHAPTER 7

IMPLEMENTATION

Design Pattern Used – Model View Controller (MVC)

The **AI Powered Placement Preparation Platform** is designed using the **Model–View–Controller (MVC)** architectural pattern to ensure proper separation of concerns, modularity, scalability, maintainability, and efficient management of system components. The MVC architecture helps organize the application into independent layers, making development, testing, debugging, and future enhancements easier and more efficient. By separating the user interface, business logic, and database operations, the system achieves better code reusability, flexibility, and overall performance.

The architecture consists of three major components: **Model**, **View**, and **Controller**. Each layer performs specific responsibilities and communicates with other layers in a structured manner to ensure smooth functioning of the application.

Model

The **Model layer** manages all data-related operations and interactions with the database. MongoDB is used as the database, and Mongoose schemas are used to define structured data models such as:

- User
- Resume
- Test
- Question
- Result
- InterviewSession
- Feedback
- Notification
- PerformanceAnalytics

Each schema enforces validation rules, maintains relationships, and ensures data consistency throughout the system.

View

The **View layer** represents the frontend user interface developed using React.js. It includes reusable components and interactive pages such as:

- Login & Registration
- Dashboard
- Roadmap Generator
- Aptitude & Coding Test Interface
- Resume Upload & Analysis
- AI Mock Interview
- Analytics Dashboard
- Notifications

React.js provides a dynamic, responsive, and user-friendly interface with real-time updates and seamless navigation.

Controller

The **Controller layer** acts as an intermediary between the frontend and backend. controller handle:

- Authentication and authorization
- Resume analysis requests
- Test management and evaluation
- AI interview processing
- Performance analytics
- Notification handling
- Placement readiness score generation

Each API request is validated, processed, and securely connected to the database and AI modules.

Frontend Implementation

The frontend of the **AI Powered Placement Preparation Platform** is developed using React.js with a component-based architecture.

Technologies Used

- React.js
- React Router
- Axios
- CSS / Tailwind CSS
- Socket.io Client

Key Features

- Modular and reusable components
- Client-side routing using React Router
- Responsive user interface
- Real-time dashboard updates
- Interactive performance analytics
- AI-powered personalized recommendations

Major Pages Implemented

- Landing Page – Introduces the platform and its features
- Authentication Pages – Login and Registration
- Dashboard – Displays performance metrics and analytics
- Roadmap Page – Personalized AI-generated preparation roadmap
- Aptitude & Coding Test Page – Conducts adaptive assessments
- Resume Analysis Page – Upload and analyze resumes
- AI Mock Interview Page – Simulates interview sessions
- Analytics Page – Displays placement readiness reports and statistics
- Notification Page – Displays updates and alerts

Backend Implementation

The backend is implemented using Node.js and Express.js following RESTful API architecture principles.

Technologies Used

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT Authentication
- Python AI Modules
- Socket.io

Key Backend Features

- RESTful API architecture
- Secure JWT-based authentication
- Role-based authorization
- AI service integration
- Modular routing and controllers
- Real-time communication support

Module-wise Detailed Implementation

1. User Authentication Module

This module manages secure user registration and login.

Features Implemented

- Secure registration and login
- JWT token generation
- Role-based access (Student, Faculty, TPO)
- Session management

Implementation Details

- Passwords are hashed before storing in MongoDB
- JWT tokens are generated after successful login
- Middleware validates authentication and access permissions

2. User Profile Management Module

This module allows users to manage academic and professional information.

Features Implemented

- Profile creation and editing
- Academic details management
- Skills and career interests management
- Profile completion tracking

Implementation Details

- Profile information is stored using Mongoose schemas
- Dynamic profile updates are handled through REST APIs
- Progress tracking is calculated automatically

3. Personalized Roadmap Generation Module

This module generates AI-based preparation roadmaps for students.

Features Implemented

- Personalized preparation plans
- Skill gap analysis
- Dynamic learning recommendations

Implementation Details

- AI analyzes academic details, test scores, and skill levels
- Machine learning algorithms generate customized learning paths
- Recommendations update automatically based on progress

4. Aptitude & Coding Test Module

This module handles placement assessment activities.

Features Implemented

- Aptitude tests
- Coding assessments
- Adaptive question difficulty
- Automatic evaluation and scoring

Implementation Details

- Questions are stored in MongoDB collections
- Scores are calculated dynamically after submission
- Adaptive logic adjusts question difficulty according to performance

5. AI Resume Analysis Module

This module analyzes resumes using Artificial Intelligence and NLP techniques.

Features Implemented

- Resume upload and parsing
- Skill extraction
- Keyword analysis
- Resume improvement suggestions

Implementation Details

- NLP algorithms analyze resume content and formatting
- Resume scores are generated dynamically
- Feedback reports are stored for future reference

6. AI Mock Interview Module

This module simulates interview scenarios using AI.

Features Implemented

- AI-powered mock interviews
- Audio/video recording
- Speech-to-text conversion
- Communication and confidence analysis

Implementation Details

- Whisper AI converts speech into text
- NLP models evaluate answer quality and communication
- AI generates interview feedback and improvement suggestions

7. Placement Readiness Score Module

This module calculates overall placement preparedness.

Features Implemented

- Readiness score calculation
- Performance tracking
- Skill evaluation analytics

Implementation Details

- Scores are calculated using test performance, resume quality, and interview analysis
- Data analytics algorithms generate overall readiness metrics

8. Notification & Recommendation Module

This module provides alerts and AI-based recommendations.

Features Implemented

- Test reminders and notifications
- AI-generated recommendations
- Real-time updates

Implementation Details

- Notifications are stored in MongoDB
- Socket.io provides real-time communication
- AI recommends learning resources and improvement areas

9. Analytics Dashboard Module

This module provides detailed analytics for students, faculty, and TPOs.

Features Implemented

- Performance dashboards
- Department-level analytics
- Downloadable reports

- Placement trend visualization

Implementation Details

- Analytics are generated dynamically using stored performance data
- Dashboards display graphs, readiness scores, and improvement insights

Frontend–Backend Integration

- Axios is used for frontend-backend API communication
- JWT tokens are sent securely through request headers
- Socket.io enables real-time communication and notifications
- AI modules interact with the backend through REST APIs
- Environment variables are managed securely using `.env` files

The **AI Powered Placement Preparation Platform** successfully integrates Artificial Intelligence, modern web technologies, and real-time analytics to provide a scalable, secure, and intelligent placement preparation ecosystem. The modular architecture, AI-powered evaluation system, personalized learning roadmap, and analytics dashboard contribute to the overall efficiency and effectiveness of the platform in improving student placement readiness.

CHAPTER 8

RESULTS

1. Landing Page

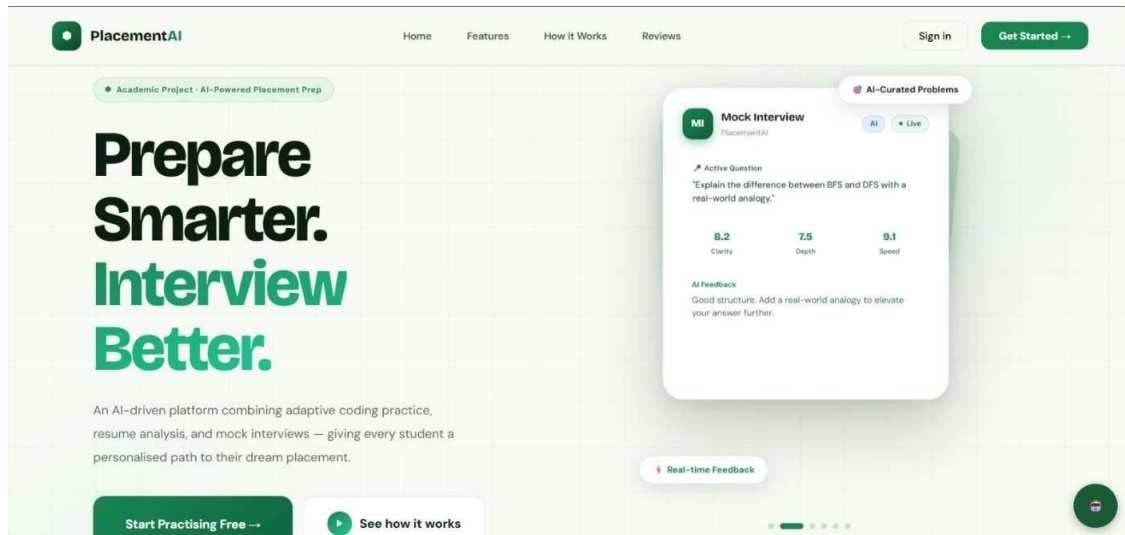


Figure 12. Landing page

2. Registration Page

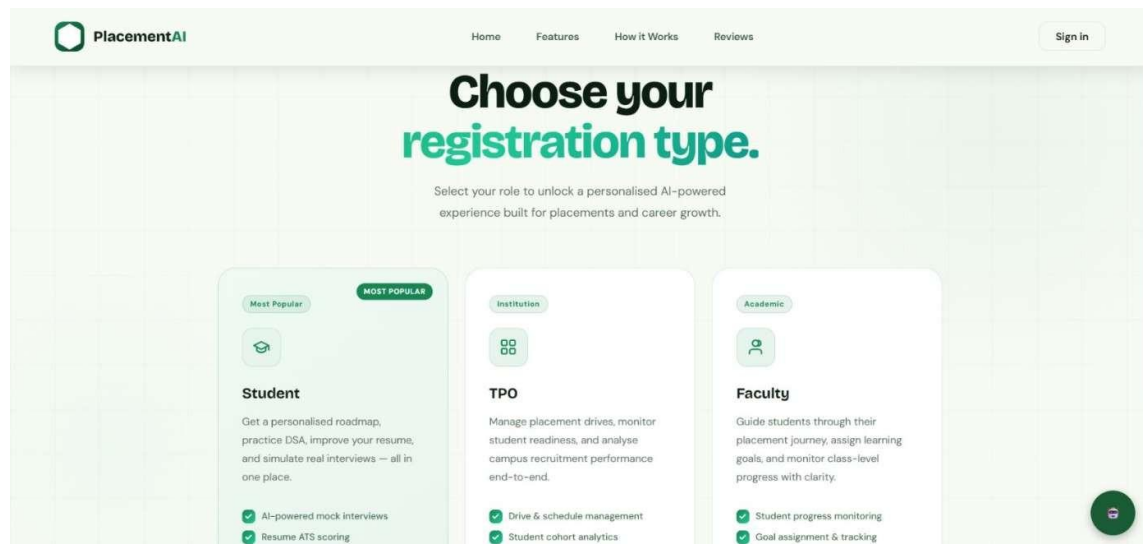


Figure 13. Registration page

3. TPO Registration Page

AI Placement Prep

INSTITUTIONAL PARTNER ACCESS

Empower Placements with Smart AI Tools

Manage student readiness, placement drives, analytics, and recruiter engagement through one intelligent dashboard.

- ✓ Student readiness tracking
- ✓ Placement drive management
- ✓ Performance analytics
- ✓ Recruiter insights

Placement Analytics

AI PLACEMENT PREP

TPO Signup

Create your Training & Placement Officer account.

Full Name: Dr. Sarah Johnson

Work Email: sarah@university.edu

College / University: Global Institute of Tech

Designation: Chief Placement Officer

Contact Number: +1 (555) 000-0000

Password: *****

Create TPO Account →

Already registered? [Login](#)

Figure 14. TPO Registration page

5. Faculty Registration Page

EDUCATOR NETWORK ACCESS

USED BY: 200+ FACULTY MEMBERS | 150+ COLLEGES | 5000+ STUDENTS

Guide Students Toward Placement Success

Join as faculty mentor to track student growth, provide guidance, and strengthen placement outcomes through AI insights.

- ✓ Student performance tracking
- ✓ Mentorship tools
- ✓ Skill gap analytics
- ✓ Placement readiness reports

AI PLACEMENT PREP

Faculty Signup

Create your faculty mentor account to begin guiding your students.

Full Name: Dr. Sarah Jenkins

Email Address: sarah.jenkins@university.edu

Password: *****

Phone Number: +1 (555) 000-0000

Department: Computer Science & Engineering

Designation: Associate Professor

Highest Qualification: PhD in Computer Science

Years of Experience: 10

Specialization: Machine Learning, Data Structures

Subjects Teaching:

Areas of Expertise:

Figure 15. Faculty Registration page

6. Student Registration Page

Figure 16. Student Registration page

7. Login Page

Figure 17. Login page

8. Student Dashboard Page

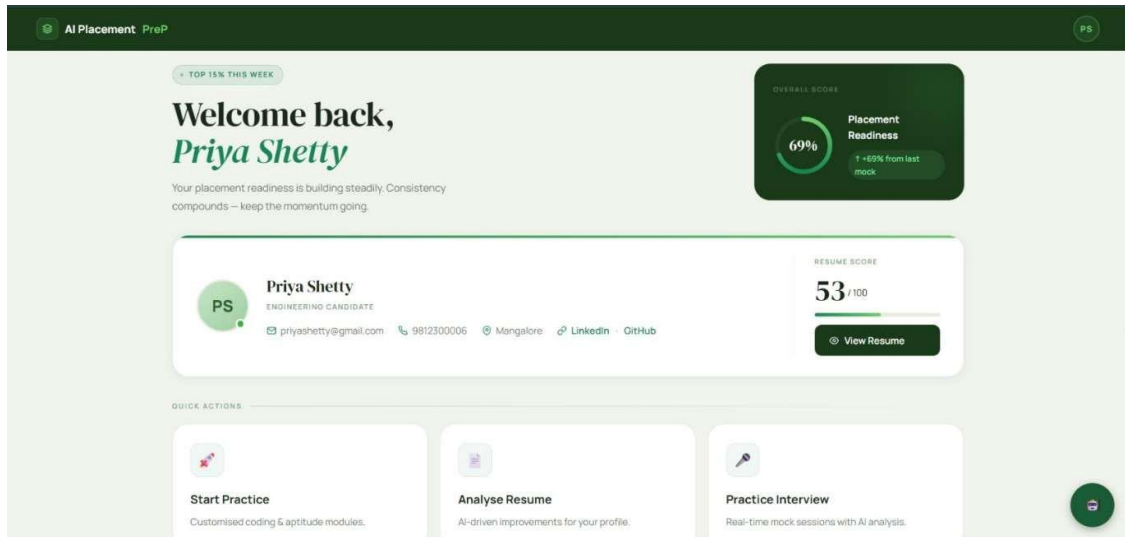


Figure 18. Student Dashboard page

9. Faculty Dashboard Page



Figure 19. Faculty dashboard page

10. TPO Dashboard Page

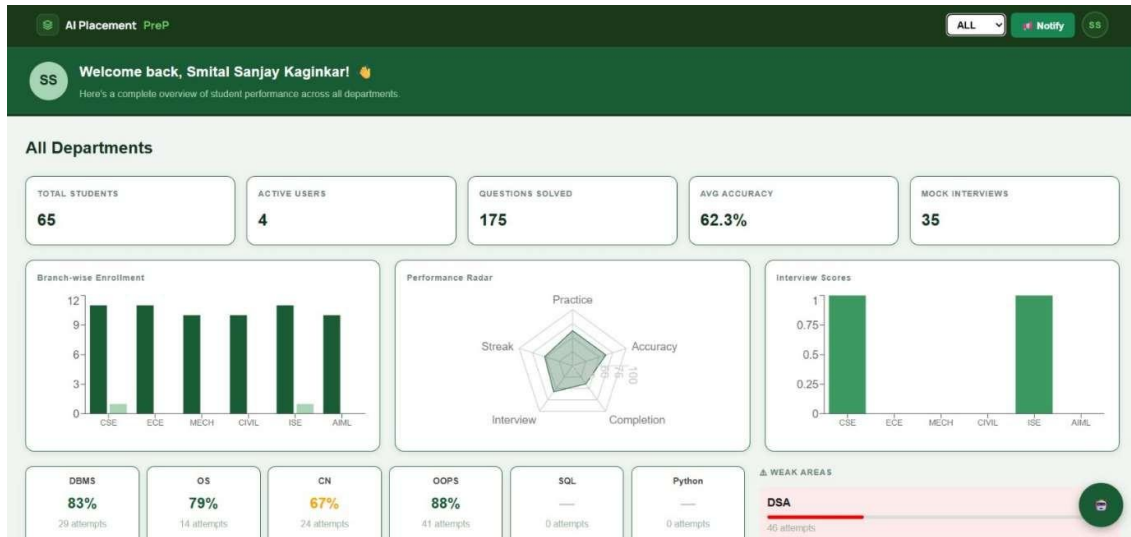


Figure 20. TPO dashboard page

11. Practice Dashboard Page

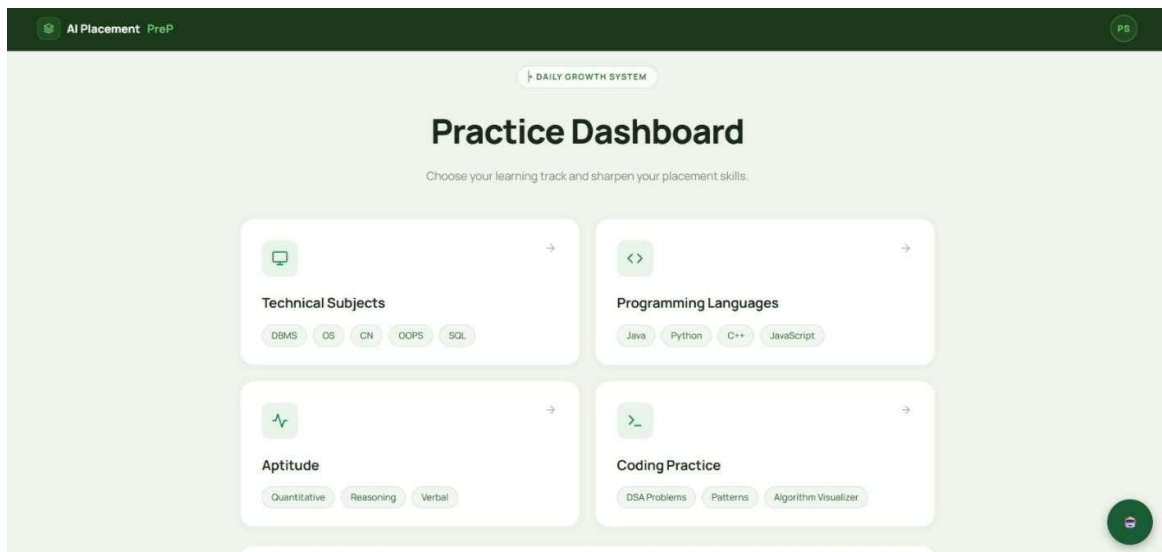


Figure 21. Practice Dashboard page

12. Aptitude Dashboard Page

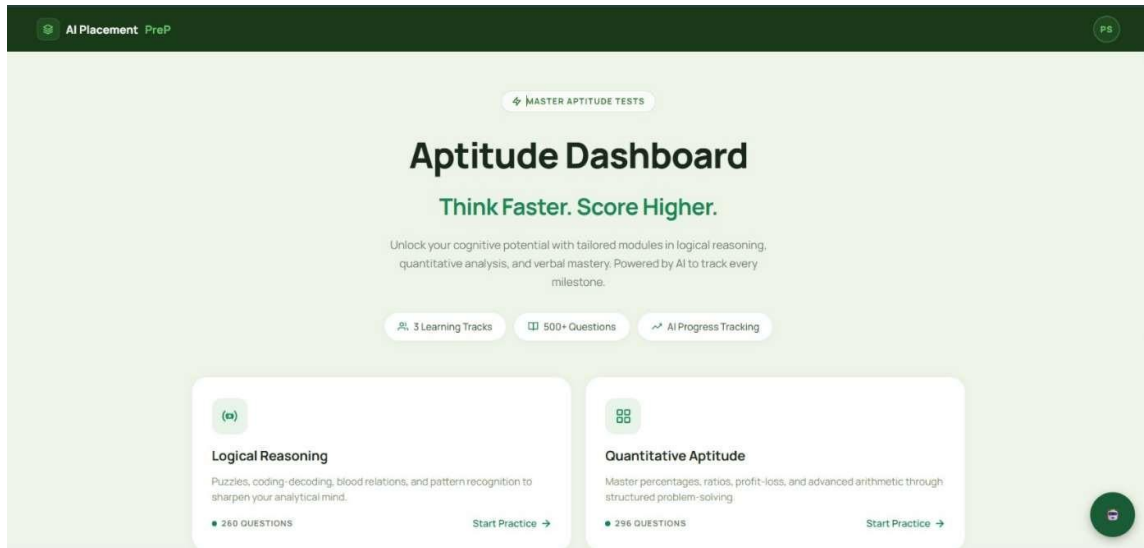


Figure 22. Aptitude Dashboard page

13. Aptitude practice Page

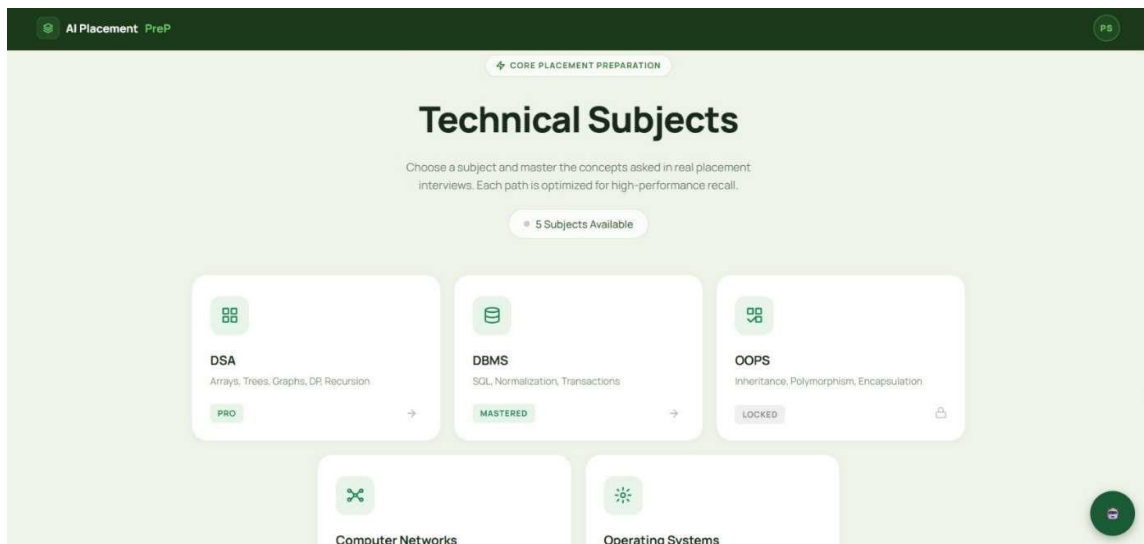


Figure 23. Aptitude Practice page

14. Aptitude practice questions Page

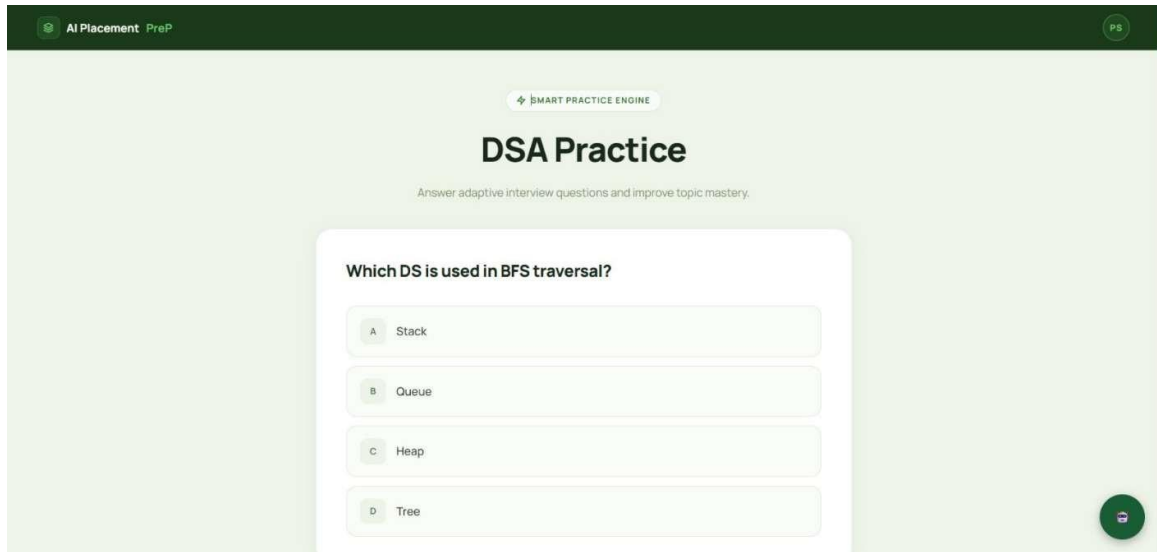


Figure 24. Aptitude Practice questions page

15. Programming Languages practice Page

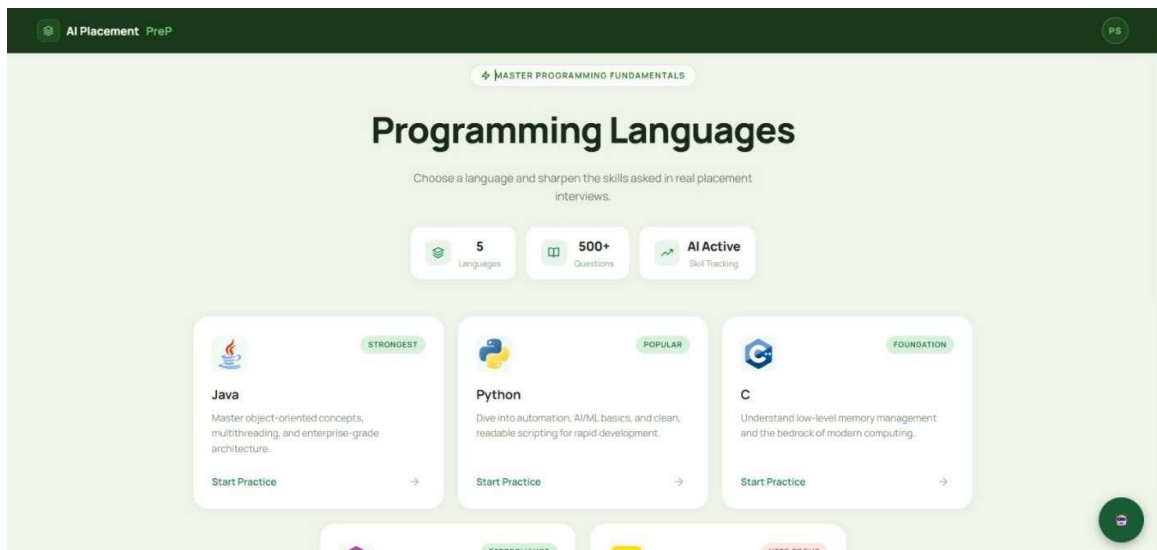


Figure 25. Programming Languages Practice page

16. Selection of coding topic Page

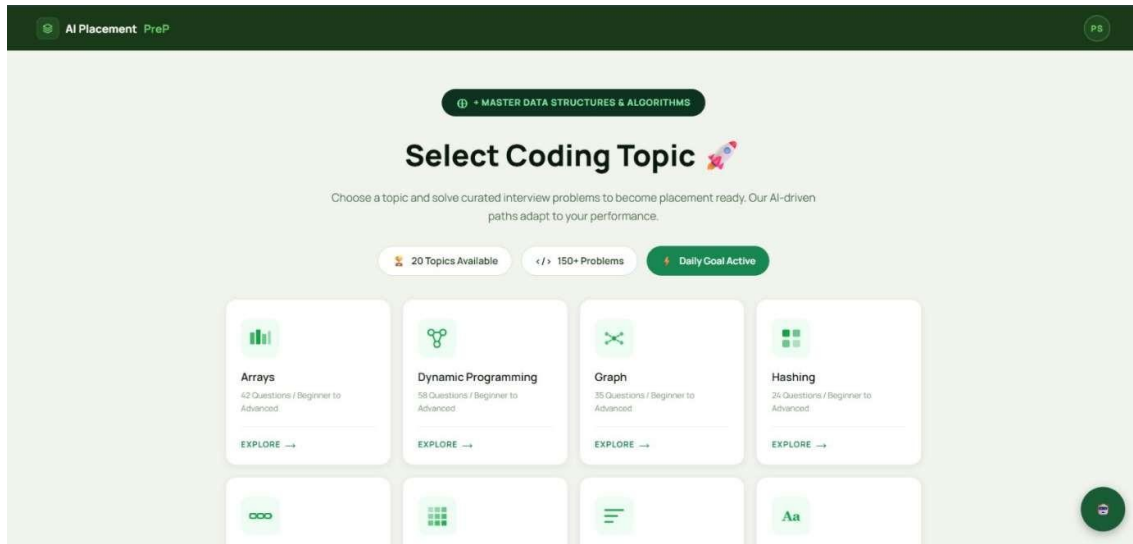


Figure 26. Selection of coding topic page

17. Coding practice Page

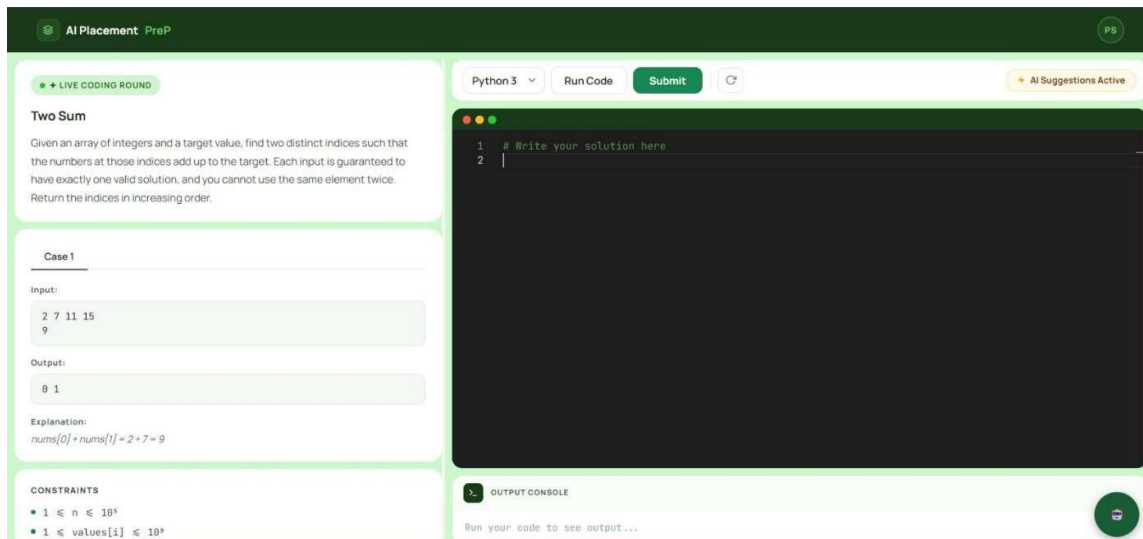


Figure 27. Coding practice page

18. Resume Analyzer Page

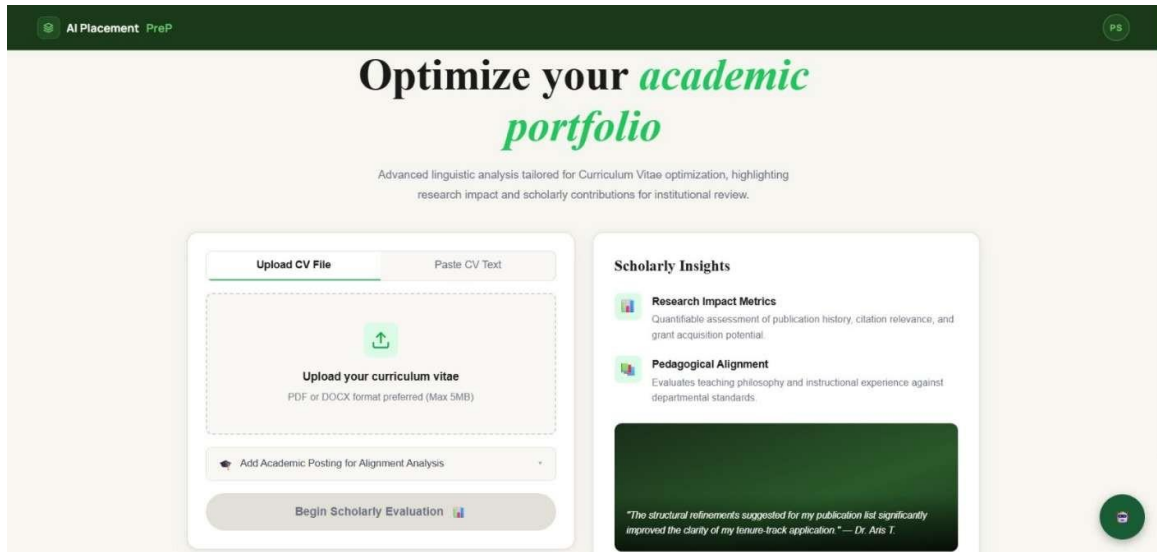


Figure 28. Resume Analyzer page

19. Mock Interview Dashboard Page

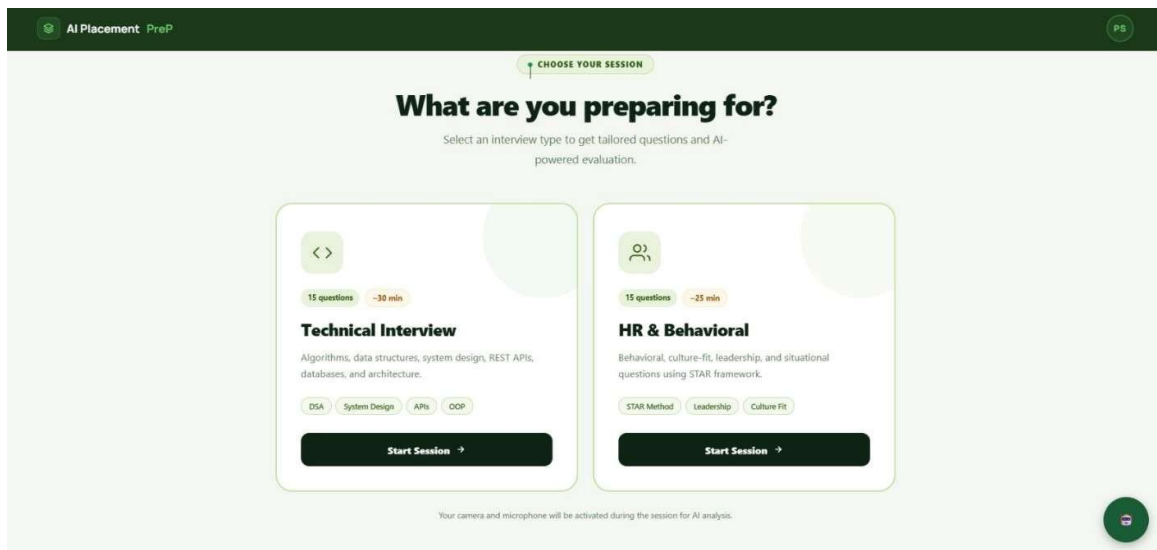


Figure 29. Mock Interview Dashboard page

20. Mock Interview Page

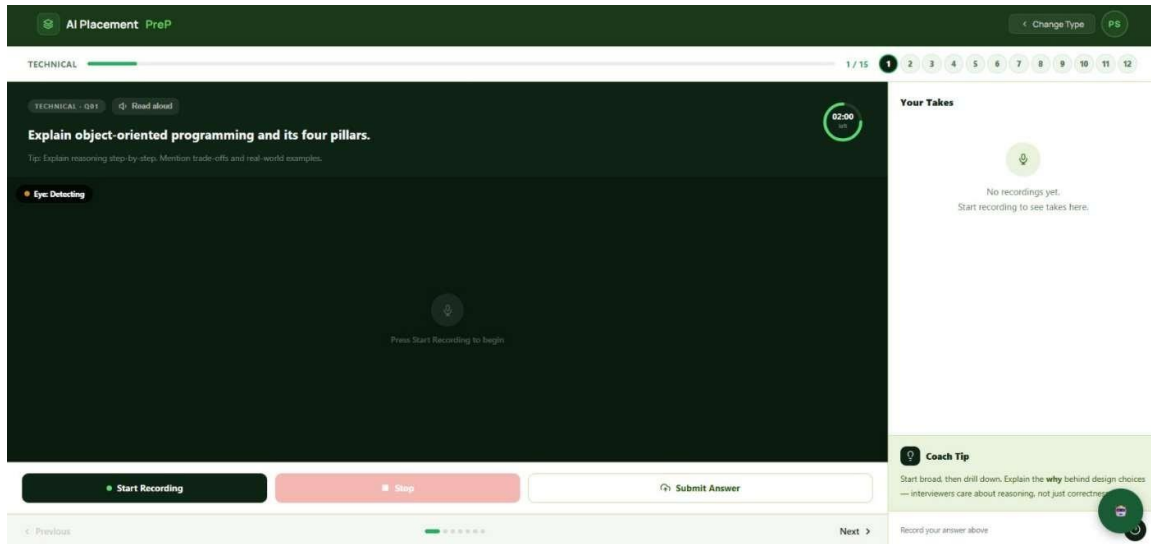


Figure 30. Mock Interview page

21. View Performance Page

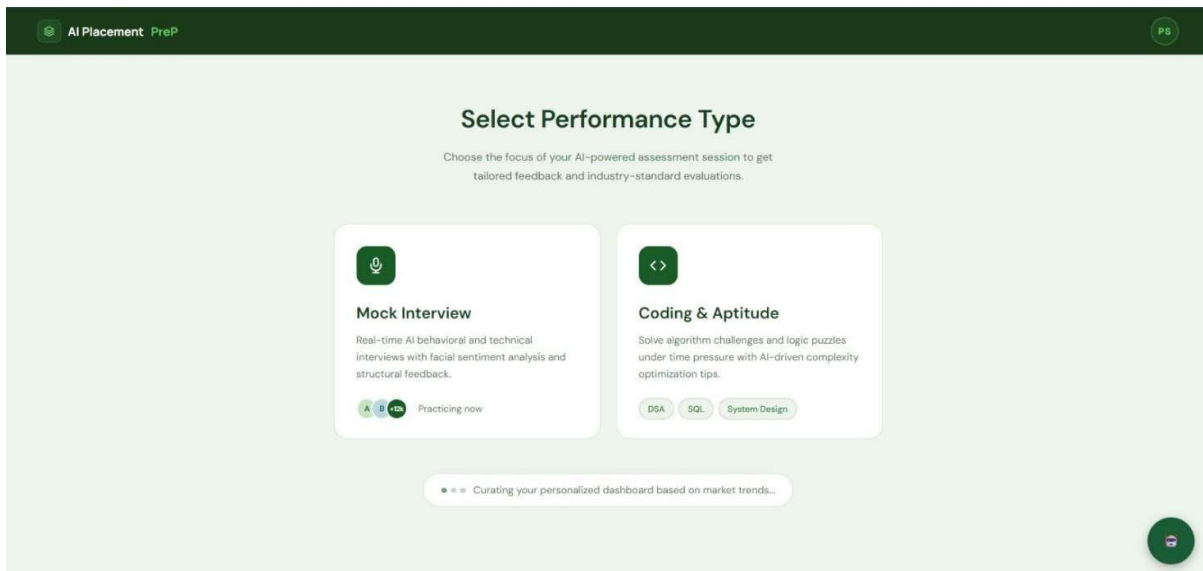


Figure 31. View Performance page

22. Mock Interview Performance Page

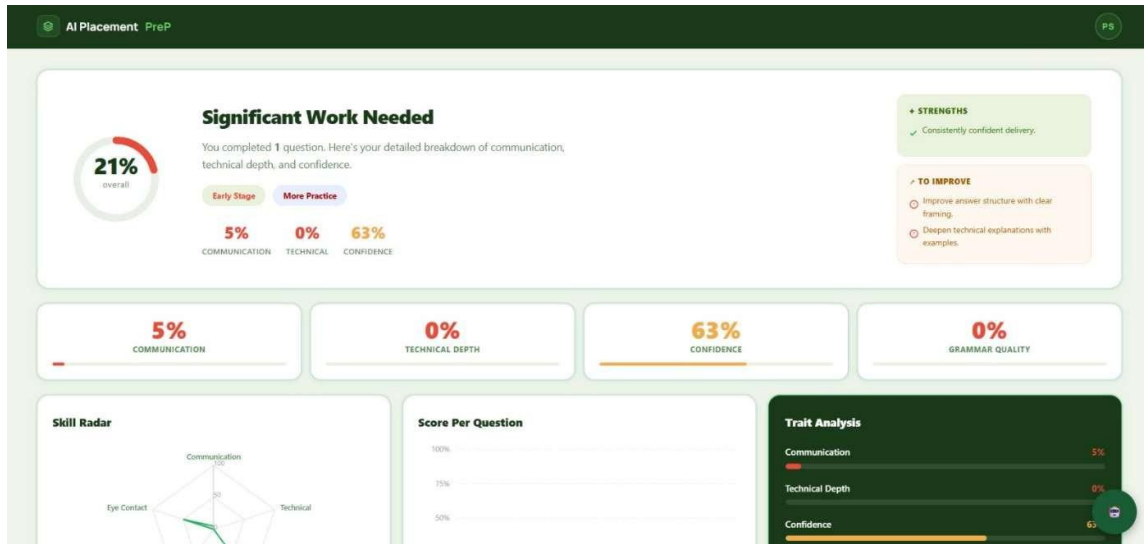


Figure 32. Mock Interview Performance page

23. Coding Performance Page

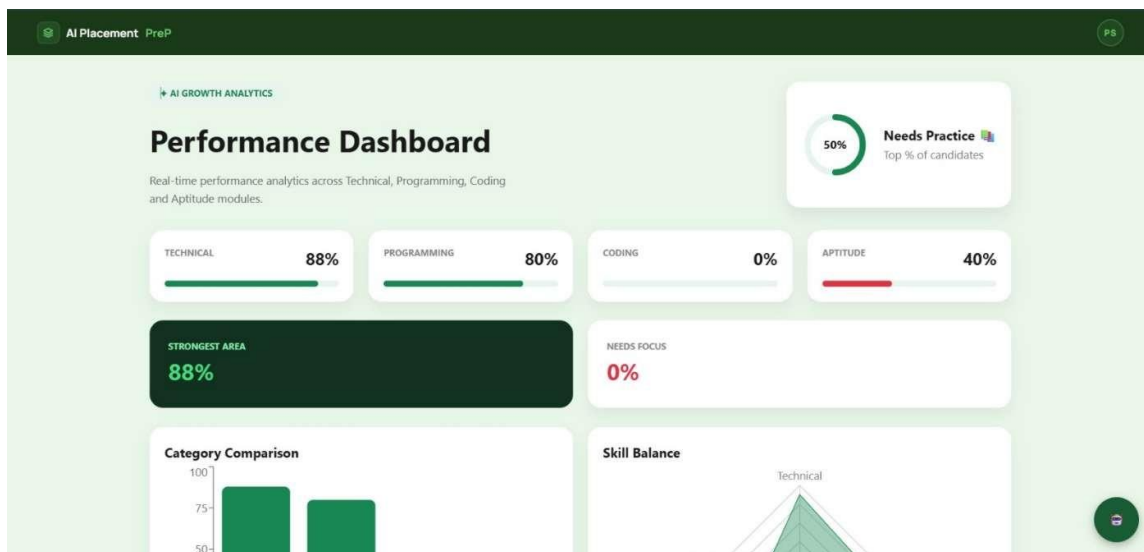


Figure 33. Coding Performance page

24. Weak Topic Analyze Page

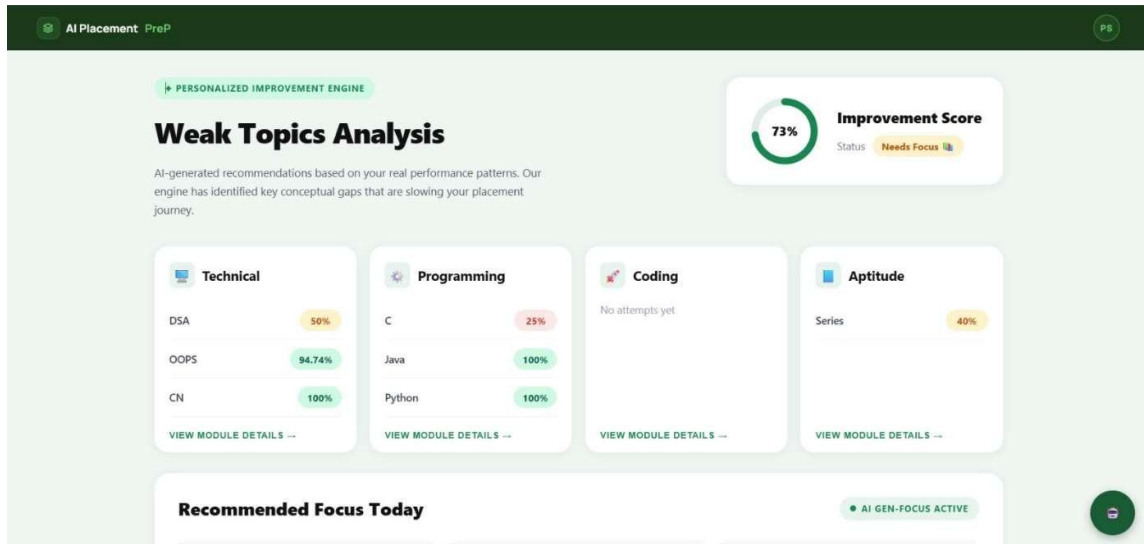


Figure 34. Weak Topic Analyze page

25. Chatbot Module

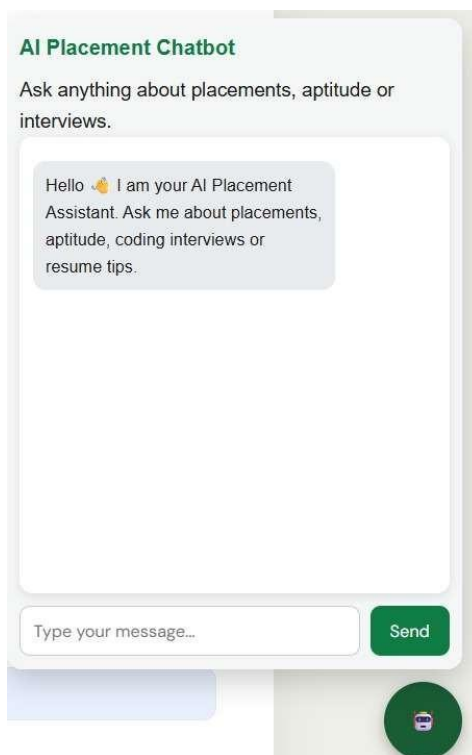


Figure 35. Chatbot Module

CHAPTER 9

CONCLUSION

The **AI Powered Placement Preparation Platform** has been successfully designed and implemented as an intelligent, secure, and user-centric system that enhances the placement preparation process for students. The project effectively addresses the challenges associated with fragmented preparation resources, lack of personalized guidance, and limited performance analysis by providing a centralized platform integrated with Artificial Intelligence technologies.

The system incorporates major functionalities such as secure user authentication, personalized roadmap generation, aptitude and coding assessments, AI-based resume analysis, AI-powered mock interviews, placement readiness scoring, real-time notifications, and performance analytics. The integration of modern technologies such as the MERN stack, Python-based AI modules, Natural Language Processing (NLP), and real-time communication technologies ensures efficient frontend–backend interaction and intelligent automation throughout the platform.

The platform provides students with a structured and adaptive learning environment that continuously evaluates performance, identifies improvement areas, and generates personalized recommendations. Features such as resume evaluation, adaptive testing, and AI-driven interview analysis help students improve technical, communication, and problem-solving skills required for campus placements.

Agile development practices were followed during the project lifecycle to ensure systematic planning, efficient collaboration, and timely implementation of modules. The modular MVC-based architecture improves scalability, maintainability, and flexibility, enabling future upgrades and integration of advanced AI functionalities.

Overall, the project successfully achieves its objective of creating a comprehensive placement preparation ecosystem that bridges the gap between academic learning and industry expectations. The platform demonstrates the practical application of Artificial Intelligence, machine learning, and modern web technologies in solving real-world placement preparation challenges. Future enhancements may include advanced voice-based AI interviews, company-specific preparation modules, mobile application support, predictive analytics, and integration with professional networking platforms such as LinkedIn.

FUTURE SCOPE

Although the core functionalities of the **AI Powered Placement Preparation Platform** are fully operational, several enhancements can be incorporated in the future to further improve the platform's intelligence, scalability, and effectiveness in placement preparation:

- **Advanced AI-Based Career Guidance System** can be integrated to provide personalized career recommendations, company-specific preparation strategies, and skill development plans based on student performance, interests, and career goals.
- **Voice and Emotion-Based AI Interview Analysis** can be implemented to analyze facial expressions, confidence levels, speaking fluency, tone, and emotional behavior during mock interviews for more realistic interview evaluation.
- **Company-Specific Placement Preparation Modules** can be added to provide customized aptitude questions, coding challenges, interview patterns, and preparation roadmaps based on recruitment processes of specific companies.
- **Mobile Application Development** for Android and iOS platforms can be introduced to provide students with seamless access to tests, interviews, analytics, and notifications from mobile devices.
- **Advanced Analytics and Predictive Placement Insights** can be incorporated to predict placement readiness, identify weak areas using machine learning models, and generate detailed performance trends for students and institutions.
- **Integration with Professional Platforms** such as LinkedIn, GitHub, and coding platforms can be implemented to automatically analyze profiles, coding activity, and professional achievements for enhanced recommendations.
- **Gamification Features** such as leaderboards, badges, achievement levels, and reward systems can be introduced to increase student engagement and motivation during preparation.
- **Real-Time Collaborative Learning Features** including peer discussions, coding contests, group mock interviews, and mentor interaction modules can further improve collaborative learning experiences.
- **Cloud-Based Scalability and Deployment** can be enhanced to support large-scale institutional adoption with improved performance, availability, and distributed AI processing.

These future enhancements will transform the **AI Powered Placement Preparation Platform** into a more intelligent, adaptive, scalable, and industry-oriented ecosystem that significantly improves student employability and placement success rates.

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- [6] **Imperious: A Social Networking Platform for Collaborative Learning and Professional Growth**, *International Journal of Research in Advent Technology*, March 2025. [Online]. Available: <https://www.ijraset.com/best-journal/a-social-networking-platform-for-collaborative-learning-and-professional-growth> **(education tech / academic social networks)**

APPENDIX A – PUBLISHED RESEARCH PAPER

AI Powered Placement Preparation Platform for Student Career Readiness

Abstract

Objectives: To develop an AI-powered platform that enhances student placement readiness through personalized learning, performance analytics, and interview simulation. **Methods:** A web-based system was developed using React.js for frontend, FastAPI for backend, and MongoDB for database management. Machine learning techniques were used to analyze student performance data and generate recommendations, while OpenAI Whisper was used for speech-to-text processing and MediaPipe for facial and gesture analysis during mock interviews. **Findings:** The system improved student performance by approximately 30% in mock assessments and reduced preparation time by 25%. Students were able to identify weak areas more effectively and showed increased engagement through personalized recommendations and interactive modules. **Novelty:** The platform integrates AI-based recommendation systems, coding evaluation, resume analysis, and voice-based interview simulation into a single unified system for placement preparation.

Keywords:

Artificial Intelligence, Placement Preparation, Career Readiness, Machine Learning, Student Analytics

1. Introduction

Campus placements are a critical phase in a student's academic journey, yet many students struggle due to lack of structured preparation, personalized guidance, and real-time feedback. Existing platforms often provide generalized content that does not adapt to individual learning needs or offer integrated support for interview preparation and performance analytics. Recent advancements in artificial intelligence and machine learning have enabled the development of adaptive learning systems that analyze user performance and provide personalized recommendations to improve learning outcomes [1]. Additionally, studies on student performance prediction using machine learning techniques have demonstrated the effectiveness of data-driven approaches in identifying skill gaps and improving academic outcomes [3]. Recommendation systems in education have also been widely explored to enhance personalized learning experiences and optimize student engagement [14].

Despite these advancements, current solutions are often limited to isolated functionalities such as content recommendation or performance tracking, and lack a unified framework that integrates assessment, recommendation, resume analysis, and interview simulation [12]. Furthermore, most existing systems do not incorporate real-time interaction and behavioral analysis during interview preparation, which is essential for improving communication skills and confidence.

To address these limitations, this work proposes an AI-powered placement preparation platform that integrates personalized learning, coding practice, resume analysis, and real-time interview simulation into a single unified system. The proposed approach aims to provide a comprehensive and adaptive learning environment that enhances student readiness and employability.

2. Methodology

The proposed system is designed as an AI-powered placement preparation platform that integrates assessment, recommendation, interview simulation, and performance analytics.

2.1 Data Collection

The dataset used includes publicly available datasets from Kaggle (<https://www.kaggle.com>) and user-generated data collected through the platform.

User-generated data is collected through the platform and includes:

- Aptitude test scores
- Coding challenge performance
- Resume content uploaded by users
- Mock interview responses
- User activity and engagement metrics

In addition to this, the platform utilizes **practice datasets and question banks** obtained from publicly available sources such as Kaggle and standard placement preparation repositories. These include:

- Aptitude and logical reasoning question sets
- Programming problems and coding challenges
- Technical multiple-choice questions (OOPS, DBMS, OS, CN)

These datasets are used to design assessment modules and provide a diverse set of questions for evaluation. The combined dataset enables the system to perform performance analysis, generate personalized recommendations, and improve the overall learning experience.

2.2 System Workflow

The system follows a structured workflow to provide personalized placement preparation:

1. Student registration and profile creation
2. Initial diagnostic assessment
3. Data preprocessing and feature extraction
4. Personalized recommendation generation
5. Continuous practice through tests and coding challenges
6. Performance tracking and feedback

This workflow enables adaptive learning by continuously analyzing student performance and updating recommendations accordingly.

2.3 Data Processing and Feature Extraction

Collected data is processed to extract meaningful features such as:

- Accuracy scores
- Topic-wise performance
- Time taken per test
- Resume keyword matching
- Interview response quality

These features are used to identify weak and strong areas of the student.

2.4 Recommendation Engine

The recommendation system analyzes user performance and generates personalized suggestions.

The system uses rule-based and machine learning logic to:

- Identify weak subjects
- Suggest relevant topics and tests
- Adjust difficulty level dynamically

This enables targeted learning and efficient preparation.

2.5 Machine Learning Model

A machine learning approach is used for performance analysis and readiness prediction.

- **Input:** Student performance metrics (test scores, activity, resume features)
- **Output:** Readiness score and personalized recommendations

Algorithms such as classification and clustering techniques are applied to group students based on skill levels and predict placement readiness. The dataset was divided into **training (80%) and testing (20%) sets** for performance evaluation to ensure the reliability and generalization of the model. Additionally, for the resume analysis module, text-based features were extracted and evaluated using similar training and testing procedures to ensure accurate feedback generation.

2.6 Resume Analysis Module

The resume analysis module evaluates student resumes using Natural Language Processing (NLP) techniques. The uploaded resumes are processed to extract key features such as:

- Skills and technologies mentioned
- Education details
- Projects and experience
- Keywords relevant to job roles

The system compares extracted keywords with industry requirements and generates feedback on:

- Missing skills
- Resume structure and formatting
- Keyword optimization

This helps students improve resume quality and align their profiles with placement requirements.

2.7 Mock Interview Module

The interview simulation module integrates:

- **Whisper:** For speech-to-text conversion
- **MediaPipe:** For facial expression and gesture analysis

This module enables real-time interview simulation and behavioral evaluation.

2.8 Evaluation Metrics

The system performance is evaluated using the following metrics:

- Accuracy of test results
- Improvement in scores over time
- Reduction in preparation time
- User engagement level
- Readiness score

2.9 System Implementation

The system is implemented using:

- Frontend: React.js
- Backend: FastAPI
- Database: MongoDB
- AI Tools: Whisper, MediaPipe
- Machine Learning: Python, Scikit-learn

3. Results and Discussion

3.1 System Architecture

The overall architecture and workflow of the proposed system consist of multiple interconnected modules designed to provide personalized placement preparation.

The system follows a client-server architecture. The frontend is developed using React.js and communicates with the FastAPI backend through REST APIs. The backend processes user inputs, interacts with the MongoDB database, and performs data analysis using machine learning techniques.

The architecture includes the following components:

- Frontend Layer: User interface for students to access tests, dashboard, and interview modules
- Backend Layer: Handles business logic, API requests, and data processing
- Database Layer: Stores user data, test scores, and performance metrics
- AI Module: Performs recommendation generation and readiness analysis
- Interview Module: Uses Whisper for speech processing and MediaPipe for facial analysis

This modular architecture ensures scalability, flexibility, and efficient data processing, similar to modern AI-based learning systems [1].

As shown in Figure 1, the system begins with user registration and profile setup, followed by adaptive testing, resume analysis, and AI-based interview simulation, ultimately generating a structured feedback report and placement readiness score.

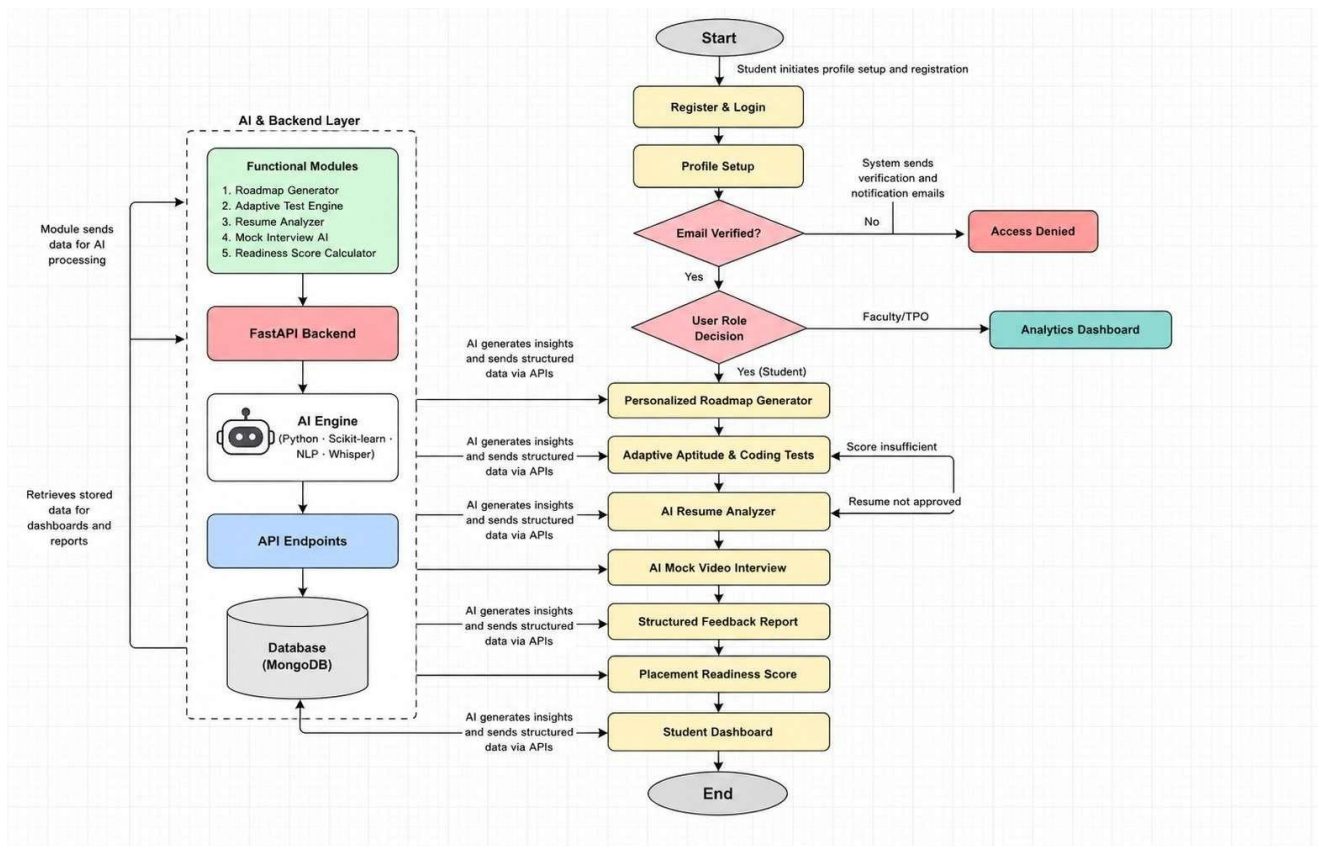


Figure 1. System architecture and workflow of the AI-powered placement preparation platform

3.2 Comparison with Existing Systems

Table 1. Comparative analysis of proposed system and traditional placement platforms

Feature	Traditional Platforms	Proposed System
Personalized Learning	Not available	AI-based recommendations
Mock Interviews	Limited	Voice + AI-based
Resume Analysis	Manual	Automated NLP-based
Performance Tracking	Basic	Advanced analytics dashboard
Coding Practice	Limited	Integrated coding module
Adaptive Difficulty	No	Yes
Real-time Feedback	No	Yes
Weak Area Detection	Manual	Automated

The comparison clearly demonstrates that the proposed system provides a more comprehensive and intelligent solution compared to existing platforms. Unlike traditional systems and existing adaptive learning approaches [14], the proposed model integrates multiple AI-driven modules such as resume analysis and interview simulation into a unified framework.

3.3 Performance Evaluation

Table 2. Performance evaluation of the proposed placement preparation system

Metric	Before System	After Using System
Mock Test Score	55%	72%
Preparation Time	High	Reduced by 25%
Weak Topic Identification	Manual	Automated
Interview Confidence	Low	Improved
Engagement Level	Moderate	High

The performance results indicate a significant improvement in student outcomes after using the proposed system. The AI-based recommendation system helped students focus on weak areas, leading to better performance and reduced preparation time. These findings are consistent with previous studies on student performance prediction and adaptive learning systems [3], which highlight the effectiveness of data-driven learning approaches.

3.4 System Performance Analysis

The dashboard provides real-time analytics, enabling students to track their performance and improvement trends. Students using the platform showed approximately 30% improvement in mock test scores, validating the effectiveness of the AI-based recommendation system. Similar improvements have been observed in AI-based educational systems that utilize personalized learning mechanisms [11].

The resume analyzer module provided structured feedback, improving resume quality and helping students align with industry expectations. Additionally, coding challenges enhanced problem-solving abilities and technical readiness, which aligns with existing research on skill-based learning platforms [10].

3.5 Interview Simulation Analysis

The mock interview module integrates Whisper for speech recognition and MediaPipe for facial and gesture analysis. This enables real-time interaction and behavioral evaluation. Unlike conventional interview preparation methods, which lack real-time feedback, AI-based interview systems provide enhanced user engagement and performance improvement [12].

Students reported improved confidence and communication skills after repeated interview practice. The system provides a realistic simulation environment, bridging the gap between theoretical preparation and real-world interviews.

4. Discussion

The proposed system demonstrates significant improvement over traditional placement preparation methods, as evidenced by the results presented in Table 2. The integration of machine learning-based recommendation systems enables personalized learning paths tailored to individual student performance, resulting in an observed improvement of approximately 30% in mock test scores.

Compared to existing platforms that primarily provide static content, the proposed system introduces dynamic and adaptive learning mechanisms. Compared to existing AI-based learning systems [1], the proposed system provides additional modules such as resume analysis and interview simulation, resulting in improved performance. The recommendation engine continuously analyzes user performance data to identify weak areas and suggest targeted practice, thereby reducing preparation time by nearly 25%.

The inclusion of coding challenges, resume analysis, and interview simulations ensures a comprehensive and holistic preparation approach. In particular, the mock interview module, powered by Whisper and MediaPipe, provides real-time interaction and behavioral analysis, enhancing communication skills and interview confidence.

Furthermore, the system provides continuous feedback through an analytics dashboard, enabling students to monitor their progress and make data-driven improvements. This iterative feedback mechanism significantly improves engagement levels and learning efficiency.

The key novelty of the proposed system lies in the integration of AI-based recommendation, real-time interview simulation, and performance analytics into a unified platform. Unlike existing solutions that focus on isolated features, this system delivers an end-to-end intelligent placement preparation framework.

5. Conclusion

This study presents an AI-powered placement preparation platform designed to enhance student career readiness through personalized learning and intelligent analytics. The system successfully integrates assessment modules, a recommendation engine, resume analysis, and real-time interview simulation into a unified platform.

The proposed system achieved approximately **30% improvement in mock test performance** and **reduced preparation time by 25%** compared to traditional placement preparation methods. These results demonstrate the effectiveness of AI-based personalized learning and adaptive feedback mechanisms in improving student outcomes. Unlike existing platforms that provide isolated features, the proposed system offers a comprehensive and integrated solution for placement preparation.

The findings indicate improved student engagement, better identification of weak areas, and enhanced interview readiness. However, the system currently relies on limited datasets and predefined evaluation metrics, which may affect scalability and generalization.

Future work can focus on integrating real-time recruiter matching, advanced emotion detection, and mobile application support to further enhance system functionality and usability.

6. Data Availability Statement

The dataset used in this study includes publicly available datasets from Kaggle (<https://www.kaggle.com>) along with user-generated data collected through the platform.

7. Ethical Clearance

This study does not involve human subjects, clinical trials, or sensitive personal data requiring ethical approval.

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IJST-2026-700 - Recommendation for major revisions

1 message

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To: nehanayak246@gmail.com

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We wish to confirm that we have received your manuscript on the aforementioned date. Shortly after submission, the manuscript was sent to the reviewers who are experts in the field. It was pointed out by them that the manuscript requires major revisions. The reviewed manuscript with comments and suggestions have been attached with this email for your reference. The reviewers seek your kind attention on the sections of the manuscript to improve upon by incorporating the suggestions and comments. Further, we request you to submit the manuscript with the requested changes and additions to be accepted for the next phase of publication. We will be awaiting your response with an improved version of your manuscript.

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Regards,

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Abstract

1. Introduction
2. Methodology
3. Results and Discussion
4. Conclusion
5. References

Title page file in detail:

Title of the article

- **Author names and affiliations**

1) *Names, Full affiliations* (complete address including the Department and the Institution) and emails of all authors in "Title page file" as separate file attachment; identify the corresponding author with email and phone number (cell) in it. 2) "*Acknowledgement*" section: Mention about funding agency/ data source/ clinical support/ Testing support/ sources of study materials. 3) "*Contributory statement*" section includes whichever appropriate: Conceptualization of study problem: Experimental Design: Software or algorithm: Data generation: Data analysis: Research Guidance: Original Drafting; Editing & Reviewing. Healthcare studies specially includes: Case examined: Intervention planned & executed: Follow-up studies. Contributory statement helps to showcase the intellectual contribution of authors, individually.

Contributory statement model (a representative case of 5 contributing authors):

Conceptualization of study problem: Dr.Aishwarya G., *Experimental Design*: Dr.Aishwarya G. & Kumaran G., *Software or algorithm*: Prof. Jalal Ahmad: *Data analysis*: Dr.Nageswara Rao, *Research Guidance*: Natarajan Gajendran, *Original Drafting or Writing*: Aishwarya G., *Editing & Reviewing*: Dr.Kumaran G.

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- **Title of the article**

Kindly insert line numbers

- **Abstract**

The abstract should be brief and the abstract should include the following sub-headings within:

Objectives:

Method:

Findings:

Novelty:

Keywords:

Enrich the abstract and conclusion with quantitative/comparative/numerical data in support of your findings (without comparative data or statistical values in these sections, it will be looked upon as empty statements as readers mostly read abstract and or conclusion only)

- **Introduction**

Introduction should distil out the gaps of previous work; abridge the introduction section to a para.

Omit expanding it as a Literature survey. Introduction should capture the major gaps of the existing literature and also to justify the present work; on this context, **major milestone works of recent and**

most matching ones (about 10 references from 2024 to 2020) should be highlighted along with their limitations for which your work is going to rectify. Generally, omit references older than 2018 (except in methodology) and no need to expand it as literature survey or restating the historical developments.

In other words, *introduction should be written in a polished way* stitching together the major advancement made in the area and also pin-point the gaps and mention how your work is going to address those issues. It should not be mere narration of the literature.

The citation should be as per the journal style i.e., numerically increasing order [1], [2], [3] etc (not author name, year) and not as zigzag like [1], [5], [3] and the same way to be listed under reference;;

Reference section should be prepared as per the journal style (Add DOI number/ publication URL for each reference);

Recent references should have been consulted (2025-2019); Please omit references older than 2018 (Except under 'Methodology' section)

- **Methodology**

Author should describe the type of data used and the method used in collection the data; provide input and output variables; state the data preprocessing approach , mention what percentage of data was used for training and testing of data; Author should state clearly the evaluation matrices used/ parameters studied. Elaborate the superiority of the proposed method by comparing gold standards. Provide URL for the data availability.

- **Results and Discussion**

Provide a **detailed discussion on how and why you were able to achieve better results in comparison to state of the art techniques/** already published reports. Your obtained results should be matched with previous publication by comparing and contrasting to exhibit the novelty of your work. Comparative citation of previous work is must.

(under introduction, the research gaps are identified by exhibiting the works of most related studies; under 'Results and Discussion', your results are to be compared with the results of previous works; this will exhibit the merits of your work)

- **Conclusion**

Conclusion should Highlight the novelty of your work (take home piece of new information) connecting with study objectives; indicate the limitations and future scope. **Support your claim with quantitative/numerical data.**

(under conclusion section, what major knowledge gaps still remain should also be stated; How your results advanced the present understandings when compared to those of previously published results, must be ascertained)

.....
Tit-Bits

- A desirable article will include:
- Short manuscript (5 to 6 pages excluding the figures and tables)
- Table and figures together less than 10 numbers
- References less than 25 (90% should be from 2025-2020)
- Abstract should be enriched with quantitative data
- Introduction is about a para, mainly highlights the recent (2024-2020) and most relevant findings (matching with your title) and expose their research gaps and justify for your work

- Methodology should include details of dataset used; how tested/ trained/ compared with gold standard methods; what modification you made in the methodology etc (omit basic details and elaboration with already known information)
- Results and discussion section should include how your results are superior compared to hitherto published claims. So, compare and contrast your findings by refereeing with previous publications (provide citation accordingly)
- (under Introduction section, you are finding the research gaps in the recently published works; but under results and discussion section" you are matching your results with relevant published reports' results)
- All the figures and Tables must be original and all figures should be of high resolution
- Conclusion section should mention how the objectives fulfilled by stating the novel outcomes of your work linking with objectives; enrich the section with quantitative data; express comparatively how your work is novel and better; mention the limitations of your work and future scope
- All references should be provided with DOI number or publication URL
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Neha Nayak <nehanayak246@gmail.com>

IJST-2026-700-R1 - Recommendation for minor revisions

1 message

Indian Journal of Science & Technology <indjst-journal@manuscriptcommunicator.com>

Sun, May 3, 2026 at 6:51 PM

To: nehanayak246@gmail.com

Dear Prof. Neha Nayak,

Manuscript Id: IJST-2026-700-R1**Manuscript Title:** AI-Powered Placement Preparation Platform for Student Career Readiness**Manuscript Type:** Research Article**Submitted Date:** 03 May 2026

We wish to confirm that we have received your manuscript on the aforementioned date. Shortly after submission, the manuscript was sent to the reviewers who are experts in the field. The manuscript has received positive response from the reviewers. Meanwhile, they have recommended for minor revisions. The reviewed manuscript with comments and suggestions have been attached with this email for your reference. The reviewers seek your attention on to improve by incorporating the suggestions and comments. Further, we request you to submit the manuscript with the requested changes and additions to be accepted for the next phase of publication. We will be awaiting your response with an improved version of your manuscript.

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During revision, please address to all comments made by the reviewers; Attach a rebuttal table with one side row contain different comments raised by reviewers and the other corresponding side should indicate your response to each comment. Also incorporate suitable changes in the manuscript, accordingly, as part of revision.

Regards,
Prof. Natarajan Gajendran (indjst@gmail.com)
Editor-in-Chief
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Neha Nayak <nehanayak246@gmail.com>

IJST-2026-700-R2 - Recommendation for minor revisions

Indian Journal of Science & Technology <indjst-journal@manuscriptcommunicator.com>

Sun, May 10, 2026 at
10:33 PM

To: nehanayak246@gmail.com

Dear Prof. Neha Nayak,

Manuscript Id: IJST-2026-700-R2**Manuscript Title:** AI-Powered Placement Preparation Platform for Student Career Readiness**Manuscript Type:** Research Article**Submitted Date:** 10 May 2026

We wish to confirm that we have received your manuscript on the aforementioned date. Shortly after submission, the manuscript was sent to the reviewers who are experts in the field. The manuscript has received positive response from the reviewers. Meanwhile, they have recommended for minor revisions. The reviewed manuscript with comments and suggestions have been attached with this email for your reference. The reviewers seek your attention on to improve by incorporating the suggestions and comments. Further, we request you to submit the manuscript with the requested changes and additions to be accepted for the next phase of publication. We will be awaiting your response with an improved version of your manuscript.

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Prof. Natarajan Gajendran (indjst@gmail.com)
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